

A Survey on Efficient Vision-Language-Action Models

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Abstract—Vision-Language-Action models (VLAs) represent a significant frontier in embodied intelligence, aiming to bridge digital knowledge with physical-world interaction. Despite their remarkable performance, foundational VLAs are hindered by the prohibitive computational and data demands inherent to their large-scale architectures. While a surge of recent research has focused on enhancing VLA efficiency, the field lacks a unified framework to consolidate these disparate advancements. To bridge this gap, this survey presents the first comprehensive review of Efficient Vision-Language-Action models (**Efficient VLAs**) across the entire model-training-data pipeline. Specifically, we introduce a unified taxonomy to systematically organize the disparate efforts in this domain, categorizing current techniques into three core pillars: (1) *Efficient Model Design*, focusing on efficient architectures and model compression; (2) *Efficient Training*, which reduces computational burdens during model learning; and (3) *Efficient Data Collection*, which addresses the bottlenecks in acquiring and utilizing robotic data. Through a critical review of state-of-the-art methods within this framework, this survey not only establishes a foundational reference for the community but also summarizes representative applications, delineates key challenges, and charts a roadmap for future research. We maintain a continuously updated project page to track our latest developments: <https://evla-survey.github.io/>.

Index Terms—Vision-Language-Action Models, Efficient VLAs, Embodied AI, Robotic Manipulation

1 INTRODUCTION

VISION-LANGUAGE-ACTION Models (VLAs) [1], [2], [3], [4] signify a breakthrough in the evolution of artificial intelligence, endowing interactive physical entities with the capacity for perception and decision-making. These models strategically ground the semantic reasoning of Large Language Models (LLMs) and the cross-modal perception of Vision-Language Models (VLMs) into a unified action space and exhibit unprecedented dexterity and generalizability in complex, unstructured environments. The potential of this paradigm is already manifest across diverse high-stakes domains, ranging from the real-time interaction in autonomous driving [5], [6] and industrial manufacturing [7] to the high-precision requirements of medical robotics [8] and laboratory automation [9].

A synergistic convergence of architectural innovation, sophisticated training regimens, and large-scale data curation propels the rapid maturation of the VLA paradigm. Architectural research focuses on synergizing multimodal backbones [10], [11] with diverse action decoders to facilitate seamless end-to-end mapping from perception to execution, spanning autoregressive policies [1] for sequential prediction to diffusion-based generators [2] for trajectory synthesis. Training methodologies have evolved toward large-scale cross-embodiment pre-training on hetero-

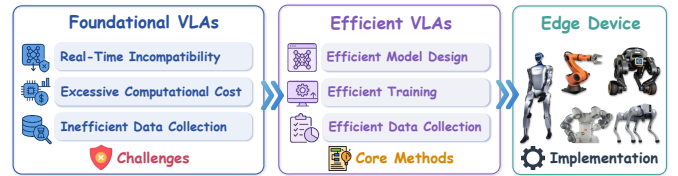


Fig. 1: **The transition from Foundational VLAs to Efficient VLAs.** Foundational VLAs are constrained by inherent challenges, specifically real-time incompatibility, excessive computational cost, and inefficient data collection. Through core methods comprising efficient model design, efficient training, and efficient data collection, efficient VLAs enable high-performance implementation on edge devices.

geneous datasets [12], progressively integrating reinforcement learning [13] and supervised fine-tuning [14] to unlock robust zero-shot generalizability and behavioral dexterity. Meanwhile, data-centric strategies prioritize the systematic acquisition of high-fidelity trajectories through expert demonstrations [15], immersive simulation [16], and generative augmentation [17] to bridge the reality gap and ensure broad generalization across diverse tasks. These multi-dimensional advancements constitute a cohesive research ecosystem that continually pushes the frontiers of autonomous agency in the physical world.

While these models have demonstrated remarkable generalist capabilities, foundational VLAs inherit the prohibitive computational and data footprints of their scaling-centric LLM and VLM foundations, resulting in a stark mismatch with the stringent latency and power constraints of physical platforms. This discrepancy creates a formidable

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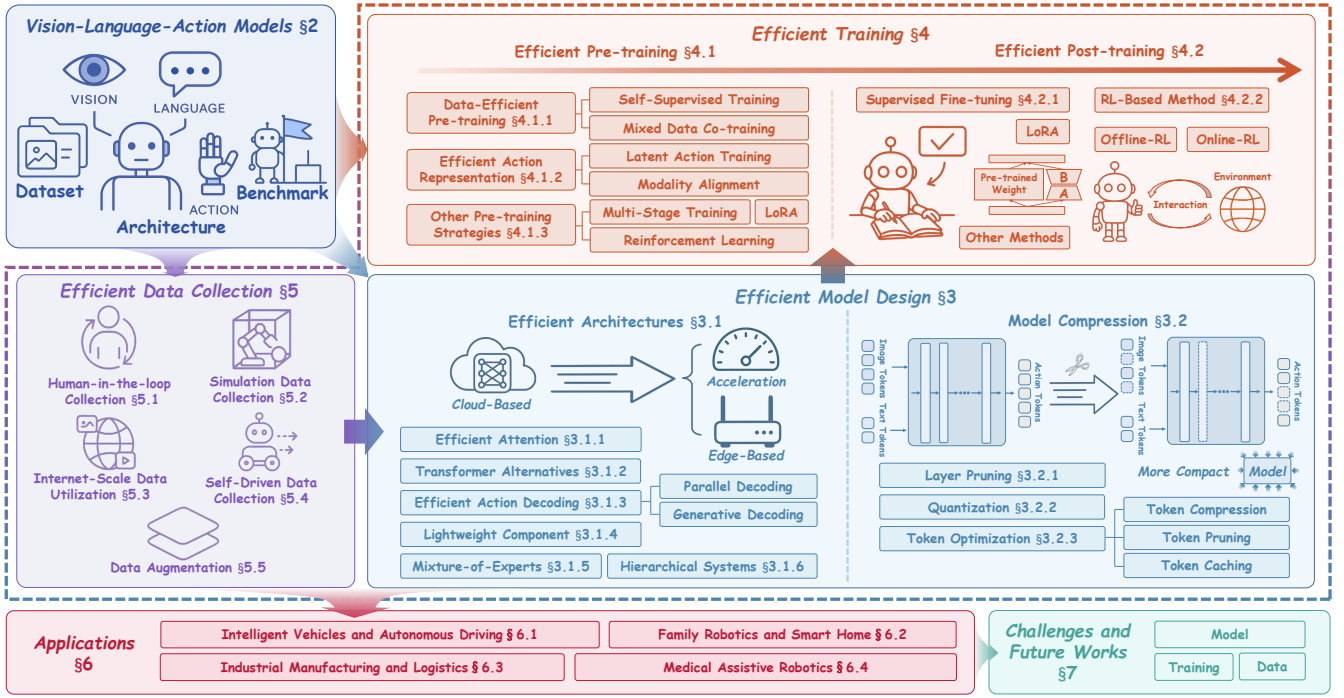


Fig. 2: **The Organization of Our Survey.** We systematically categorize efficient VLAs into three core pillars: (1) **Efficient Model Design**, encompassing efficient architectures and model compression techniques, (2) **Efficient Training**, covering efficient pre-training and post-training strategies, and (3) **Efficient Data Collection**, including efficient data collection and augmentation methods. The framework also reviews VLA foundations, key applications, challenges, and future directions, establishing the groundwork for advancing scalable embodied intelligence.

barrier to transitioning digital intelligence into real-time actuation on edge devices, such as robotic manipulators and mobile agents, manifesting primarily through the following critical bottlenecks: (1) **Real-Time Incompatibility**: current VLAs suffer from high inference latency and insufficient control frequency [1], making them incompatible with the sub-second control cycles required for responsive and adaptive robotic manipulation; (2) **Excessive Computational Cost**: the pretraining requirements are prohibitively large, as exemplified by OpenVLA [1] consuming 21,500 A100-GPU hours on a 64-GPU cluster, posing significant barriers to reproducibility and scalability; and (3) **Inefficient Data Collection**: The reliance on large-scale datasets, with examples like π_0 [2] requiring over 10,000 hours of robotic trajectories, results in a time-consuming and labor-intensive data collection pipeline that severely limits the applicability of such approaches.

To surmount these challenges, the **Efficient VLA** paradigm has emerged, encompassing resource-conscious embodied models that optimize any efficiency dimension within the model-training-data lifecycle to rectify the sub-optimal resource utilization and restricted inference speeds of foundational designs. As illustrated in Fig. 1, these designs [14], [18], [19], [20], [21], [22] bridge the gap between heavyweight foundations and edge deployment, ensuring high-frequency control and robust performance across diverse robotic scenarios.

The overarching goal of this survey is to provide a comprehensive view of the technological advances in efficient VLAs. As illustrated in Fig. 2, we organize the literature in

a taxonomy comprising three main categories: (1) **Efficient Model Design**: encompassing innovative strategies to optimize the architectural and inference efficiency of VLAs. This includes *Efficient Architectures*, which offers innovative structural optimizations alongside scalable frameworks, and *Model Compression*, which involves techniques to compress model parameters and refine token handling. (2) **Efficient Training**: incorporating advanced approaches to reduce the computational and data burdens during VLAs training, containing *Efficient Pre-training/Post-training*. The former improves efficiency through optimized data and action modeling, while the latter enhances adaptability and performance via customized learning strategies, e.g., supervised fine-tuning and reinforcement learning. (3) **Efficient Data Collection**: focusing on cutting-edge methods to improve data collection and enrichment for VLAs development, using interactive, simulated, reusable, and self-driven strategies to make dataset acquisition more scalable and efficient. Through this exploration, we aim to provide a comprehensive understanding of the current state of efficient VLAs, thereby highlighting key challenges and advancements in embodied intelligence and robotic learning.

Although several excellent surveys [23], [24], [25], [26], [27], [28] have charted the landscape of general VLAs or the broader field of embodied AI, a systematic review dedicated to the crucial aspect of efficiency in VLAs has been conspicuously absent. This work fills this critical void, uniquely consolidating the disparate research efforts focused on optimizing VLAs for real-world deployment in resource-constrained scenarios, making it an indispensable

reference for researchers and practitioners. We hope this survey together with our Github repository can help researchers and practitioners navigate through the literature and serve as a catalyst for inspiring further research on efficient VLAs.

The primary contributions of this survey are summarized as follows:

- **Pioneering Survey:** To the best of our knowledge, this work presents the first comprehensive survey specifically dedicated to the realm of efficient VLAs that covers the entire “model-training-data” process. It fills a critical gap in the literature and aims to establish a foundational reference for the research community.
- **Novel Taxonomy:** We introduce a novel and systematically structured taxonomy that organizes the core technical landscape for building efficient VLAs into three interconnected pillars: Efficient Model Design, Efficient Training, and Efficient Data Collection.
- **Future Roadmap:** We critically distill the key challenges and current limitations plaguing the field, thereby outlining promising and forward-looking research directions to inspire and guide future endeavors in scalable embodied intelligence.

The remaining sections of this survey are organized as illustrated in Fig. 2. We begin in section 2 by introducing the foundational concepts of Vision-Language-Action models, delineating the computational challenges that necessitate the pursuit of efficiency. Section 3 delves into efficient model design, systematically categorizing techniques that span efficient architectural innovations and model compression. Section 4 investigates efficient training paradigms, encompassing strategies to reduce the computational overhead and dataset demands during both pre-training and post-training stages. Section 5 examines the crucial role of efficient data collection, exploring methodologies for scalable data collection and effective augmentation to maximize data utility. We then survey key applications in section 6 where efficient VLAs have demonstrated significant potential, highlighting their practical utility in real-world scenarios. Section 7 discusses current challenges and outlines promising future research directions. Finally, section 8 summarizes the survey.

2 VISION-LANGUAGE-ACTION MODELS

This section delineates the expansive landscape of VLAs, scrutinizes the critical efficiency bottlenecks in foundational paradigms that necessitate the transition toward efficient VLAs, and underscores the pioneering contribution of this survey as a dedicated roadmap for resource-conscious embodied intelligence.

2.1 An Overview of VLAs

2.1.1 Foundational Pipeline

As depicted in Fig. 3, VLAs’ foundational pipeline divides multimodal inference into three synergistic modules: a visual encoder that distills scene images into patch embeddings; a large language model (LLM) backbone, augmented

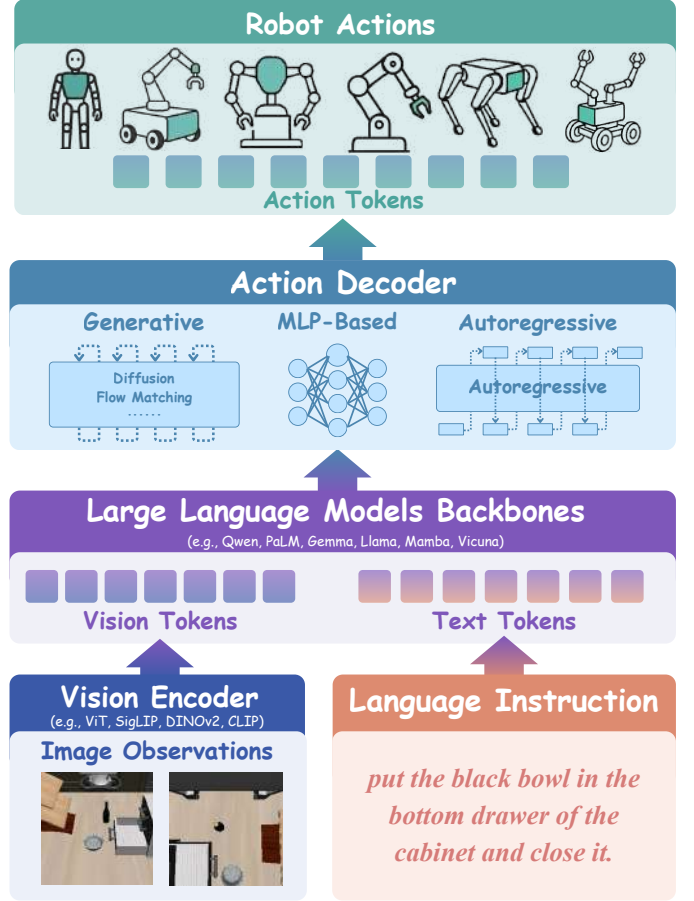


Fig. 3: **An Overview Of VLAs.** VLAs integrate vision encoders to extract visual features, LLM backbones to fuse multimodal inputs, and action decoders (MLP-based, autoregressive, or generative) to produce robotic control signals, enabling end-to-end vision-language-action reasoning for embodied manipulation tasks.

for vision-language fusion, that orchestrates high-level reasoning; and an action decoder that yields precise control trajectories. By harnessing pre-trained models, VLAs transcend task silos, fostering scalable agents adept at long-horizon manipulation.

Vision Encoder. At the ingress, the vision encoder $E_{img}(\cdot)$ ingests RGB observations and extracts their hierarchical features. Prevailing choices include vision transformers like ViT [29], SigLIP [30], DINOv2 [31], and CLIP [32], whose contrastive or self-supervised pre-training on vast corpora imparts zero-shot versatility. Abstractly, this module maps raw imagery to semantic tokens:

$$\mathbf{v} = E_{img}(I; \theta_{img}), \quad (1)$$

where $I \in \mathbb{R}^{H \times W \times 3}$ denotes the input images, $\mathbf{v} \in \mathbb{R}^{N_v \times D_v}$ represents the vision tokens, and θ_{img} indicates the encoder parameters.

LLM Backbone. After a projector $P(\cdot)$ that ensures seamless alignment and mitigates modality gaps, the multimodal sequence then feeds into the LLM backbone $LLM(\cdot)$ —the center for semantic reasoning. Spanning Qwen [33], PaLM [34], Gemma [35], Llama [36], Mamba [37], and Vicuna [38],

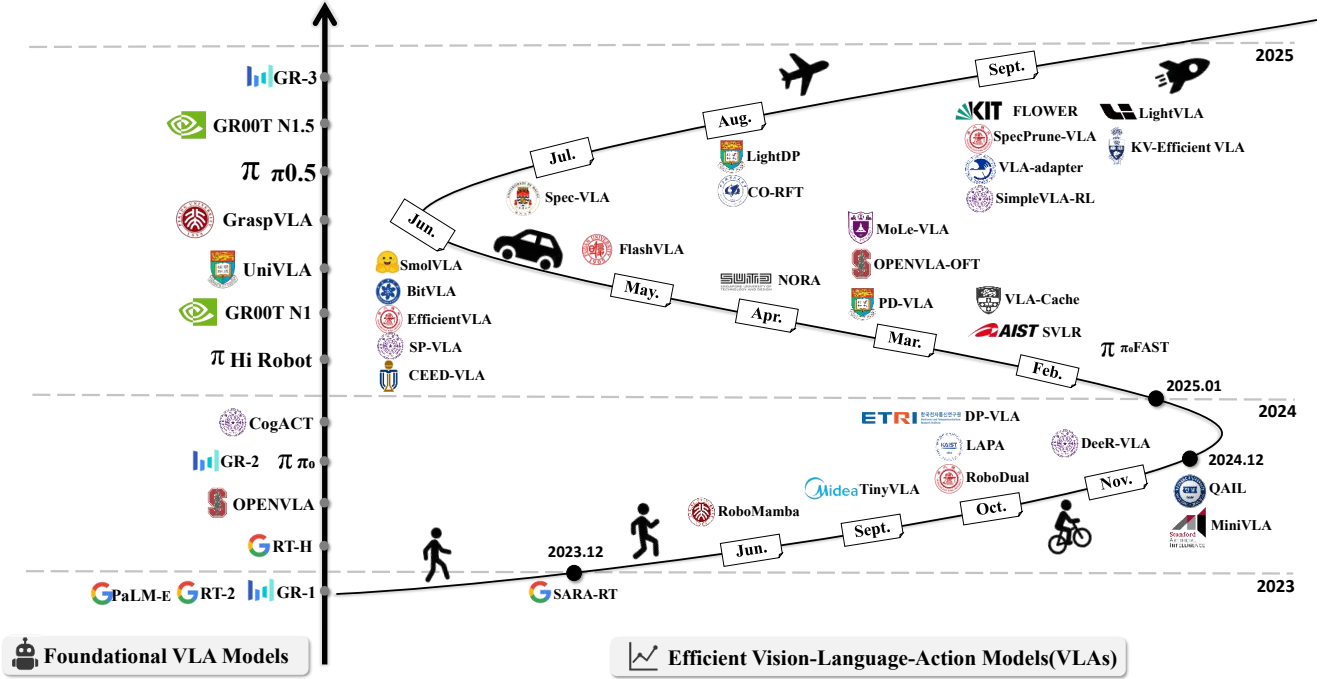


Fig. 4: **Timeline of Foundational VLA Models and Efficient VLAs.** The timeline illustrates the progression of foundational VLA models and efficient VLAs from 2023 to 2025, highlighting the explosive growth in enhancing the efficiency of VLA to bridge computational demands with real-world robotic deployment.

pre-trained LLMs process the fused embeddings for task planning, which can be formulated as follows:

$$\mathbf{h} = LLM(P(\mathbf{v}, \mathbf{l}); \theta_{LLM}), \quad (2)$$

where $\mathbf{l} \in \mathbb{R}^{N_l \times D_l}$ is language tokens, $\mathbf{h} \in \mathbb{R}^{N_h \times D}$ denotes the hidden states and θ_{LLM} represents the LLM parameters.

Action Decoder. Culminating the flow, the action decoder transmutes latents into robot-ready outputs, such as end-effector poses and gripper commands. Common instantiations include diffusion/flow matching [39], [40], [41], [42] for stochastic refinement of trajectories, autoregressive decoding for sequential action prediction, and MLP-based architectures for lightweight design. They all generate action sequences conditioned on reasoning:

$$\mathbf{a}_{1:T} = D_{act}(\mathbf{h}; \theta_{act}), \quad (3)$$

where $\mathbf{a}_{1:T} \in \mathbb{R}^{T \times D_a}$ denotes the action chunk over T timesteps and θ_{act} indicates the decoder parameters.

In essence, this tripartite architecture, including visual encoding, linguistic reasoning, and action generation, unifies perception and execution in an end-to-end framework, propelling VLAs toward generalizable embodied intelligence.

2.1.2 Datasets

The proliferation of VLAs is fundamentally underpinned by high-fidelity datasets and rigorous benchmarks, which serve respectively as the essential fuel for behavioral emergence and the critical yardstick for performance validation.

Real-world Datasets. A cornerstone of VLA training is the Open X-Embodiment (OXE [43]) dataset. This collab-

orative effort aggregates data from multiple robotics labs, encompassing a vast array of tasks, robot morphologies, and environments. Its scale and diversity are instrumental for training models with strong generalization and transfer learning capabilities. Complementing OXE are other significant datasets such as BridgeData and BridgeData V2 [44], [45], which focuses on cross-domain tasks, and DROID [46], known for its large-scale dexterous teleoperated data “in the wild”. Expanding this ecosystem, EgoDex [47] introduces high-fidelity egocentric teleoperation data to bridge the gap between human-like perception and dexterous manipulation. Simultaneously, AgiBot-World [15] provides a massive collection of 1M+ robotic trajectories, specifically curated to enhance the scalability of intelligent agents in complex, real-world industrial and service scenarios. These datasets provide the crucial “embodied” experience necessary for VLAs to learn the dynamics and constraints of the physical world.

Simulation Datasets and Benchmarks. Simulation platforms serve as the strategic engine for synthesizing large-scale, high-fidelity trajectories that circumvent the scalability constraints of physical collection. RoboGen [48] exemplifies the generative simulation paradigm by autonomously creating vast arrays of tasks and demonstrations, while RoboCasa [49] provides a large-scale environment tailored for daily household activities. To address the demand for specialized manipulation, SynGrasp-1B [50] contributes an extensive synthetic collection focusing on diverse grasping primitives, and RoboTwin [16] generates dual-arm cooperative trajectories with diverse data augmentation tricks. These platforms collectively enable the acquisition of diverse experience, facilitating the pre-training of VLAs with enhanced robustness and scalability.

Certain environments bridge the gap between data acquisition and performance validation, serving as both training resources and standardized testbeds. RLBench [51], RoboTwin [16] and Meta-World [52] are widely utilized for their predefined task libraries and consistent evaluation metrics across various manipulation scenarios. Beyond data generation, specialized benchmarks have been established to isolate and evaluate specific VLA capabilities. LIBERO [53] focuses on knowledge retention within life-long learning settings, while CALVIN [54] assesses generalization to novel instructions and long-horizon reasoning. Crucially, SIMPLER [55] provides a high-fidelity evaluation framework designed to bridge the sim-to-real gap, offering a reliable proxy for predicting the real-world performance of VLA policies. Furthermore, VLABench [56] evaluates long-horizon and multi-dimensional reasoning capabilities through massive standard tasks. In summary, these simulated ecosystems constitute a multidimensional evaluation framework that ensures the reliability and practical utility of VLAs before physical deployment.

2.2 The Need for Efficient VLAs

Despite the formidable capabilities demonstrated by VLA foundational models, their adoption beyond research prototypes faces significant barriers due to profound inefficiencies. These challenges stem from the two attributes that underpin their success: the model architecture and the data pipeline.

TABLE 1: **Efficiency-Related Metrics of Representative VLA Models.** The table compares the number of parameters, inference latency, and operating frequency of various representative VLAs, where $\downarrow$ indicates that lower values are better, and $\uparrow$ indicates that higher values are better.

Models	Params ($\downarrow$)	Infer. Latency (ms) ($\downarrow$)	Freq. (Hz) ($\uparrow$)
RT-2-PaLI-X [3]	55B	330-1000	1-3
RT-2-PaLI-X [3]	5B	200	5
Open-VLA [1]	7B	166	6
π_0 [2]	3.3B	73	20/50
HiRobot [57]	3B	73	10/50
GR00T N1 [12]	2.2B	63.9	-

From an architectural perspective, the primary impediment resides in the massive computational footprint of transformer-based VLM backbones. Characterized by billions of parameters [10], [11], [58], [59], these models incur prohibitive inference latency, resulting in control frequencies that fail to meet the stringent real-time requirements of dynamic robotic actuation, as evidenced in Tab. 1. This inefficiency is further exacerbated by the high dimensionality of visual inputs, which necessitates the intensive processing of dense token sequences through deep layers.

Regarding training methodologies, the current paradigm is burdened by the exorbitant computational budgets and energy consumption required for large-scale pre-training. Achieving robust generalization across heterogeneous embodiments demands massive compute resources, creating a significant barrier to entry and restricting the iterative development of VLAs to well-resourced institutions.

Lastly, from a data perspective, the acquisition of high-fidelity trajectories remains a critical bottleneck. Whether

through labor-intensive real-world demonstrations [43], [45] or expert-intensive simulation design [48], the curation of diverse datasets is inherently time-consuming and difficult to scale, significantly inflating the overall cost of the development lifecycle.

These profound inefficiencies collectively culminate in a critical deployment bottleneck for applications, particularly in resource-constrained settings such as autonomous vehicles and affordable consumer robotics. Therefore, the pursuit of efficiency is not merely an optional optimization but a fundamental prerequisite for unlocking the transformative potential of VLAs across the broader robotics landscape. Addressing this imperative, the field of **Efficient VLAs** has emerged. As illustrated in Fig. 4, the developmental trajectories of foundational VLAs and efficient VLAs are progressing in parallel. Notably, the timeline reveals a distinct acceleration in the emergence of efficient VLAs, particularly from late 2024 onwards. This proliferation underscores a rapidly intensifying research focus on resource-efficient models, likely driven by the demands of real-world robotic deployment.

2.3 Relevant Surveys

The burgeoning development of VLAs is establishing this framework as a cornerstone paradigm within embodied intelligence, catalyzing a surge of scholarly surveys [23], [24], [25], [26], [27], [28] dedicated to its systematic exposition. These surveys predominantly focus on the definitions, compositional constructs, model architectures, and training paradigms, as well as datasets and benchmarks pertinent to VLAs. For instance, Ma et al. [23] provides a comprehensive overview of VLA components, encompassing both low-level control policies and high-level task planners, along with a wide array of datasets and benchmarks, thus offering a holistic examination of various aspects of VLA models in embodied intelligence. Shao et al. [24], in addition to summarizing mainstream VLA architectures, introduces a perspective driven by Large Vision-Language Models (VLMs), elucidating how large-scale pre-trained VLMs have influenced the evolution of VLAs. Xiang et al. [25] draws an analogy between VLA model post-training and human motor learning, proposing a structured taxonomy aligned with human learning mechanisms. Zhong et al. [26] identifies the design of action tokenizers as the core of VLA architectural design and systematically categorizes the design principles of mainstream VLA action tokenizers. Although [23] and [24] briefly touch upon works related to efficient VLAs in certain fields, their coverage is neither thorough nor has it culminated in a universally accepted taxonomy. Consequently, the field still lacks a dedicated survey focusing on efficient VLAs. This survey aims to fill this critical gap by serving as the first comprehensive review of efficient VLAs, concentrating on the entire lifecycle across data, model, and training. It seeks to systematically dissect and synthesize the architectural, algorithmic, and optimization strategies that enable efficient VLA development and deployment, thereby laying a foundational cornerstone for future research aimed at creating scalable, resource-conscious, and practically deployable embodied AI systems.

3 EFFICIENT MODEL DESIGN

Efficient model design bridges the gap between foundational complexity and resource-constrained deployment through lightweight yet powerful architectures and advanced compression techniques that uphold semantic integrity while sharply reducing parameter counts and computational costs. This chapter introduces a unified taxonomy comprising two complementary facets of efficient model design: (1) **Efficient Architectures** that optimizes structures via modular designs, inference accelerations, and sparse processing to enhance efficiency without compromising model potency; and (2) **Model Compression** strategies that minimize representational redundancy through layer pruning, quantization, and token optimization to yield compact, performant models.

3.1 Efficient Architectures

As illustrated in Fig. 5, this subsection thoroughly examines seminal contributions, from efficient attention and transformer alternatives to decoding accelerations via parallelization and generative paradigms, lightweight components, mixture-of-experts, and hierarchical processing. All representative works on efficient architectures are summarized in Tab. 2.

3.1.1 Efficient Attention

Transformer [60] is the foundational model in modern neural architectures [29], [61], with attention mechanism at its core, ubiquitously facilitating feature alignment and multimodal fusion across domains, including VLAs where they orchestrate perceptual-linguistic synergies for embodied reasoning. However, attention’s inherent quadratic complexity in sequence length engenders prohibitive computational burdens, particularly for protracted action horizons in real-time robotics.

To address these, optimizations emerge along three axes: linear-time architectures [62] that subvert quadratic scaling, efficient masking strategies [63], [64] that prune redundant interactions, and KV-cache optimizations [65], [66] that streamline memory-bound inference.

In linear-time paradigms, SARA-RT [62] introduces an up-training regimen to seamlessly transmute Transformers into linear-attention counterparts, preserving representational fidelity to enable real-time control under constrained budgets. For efficient masking, Long-VLA [63] deploys phase-aware input masking, directing focus to static camera tokens during movement phases and gripper tokens during interactions, forging a robust mechanism for extended operations; meanwhile, dVLA [64] pioneers a unified diffusion scaffold with prefix attention masks, synergizing with KV caching to curtail inference compute and memory footprints. KV-cache refinements manifest in RetoVLA’s [65] strategic injection of discarded register tokens as auxiliary key-value pairs into action experts, augmenting cross-attention with global spatial context for efficient decision-making without core complexity escalation. Furthermore, KV-Efficient VLA [66] compresses historical KV caches into informative chunked representations via lightweight recurrent gating, adaptively retaining salient contexts for streamlined autoregressive flows.

Collectively, these refinements distill attention’s essence into scalable VLA pipelines, balancing expressiveness with economical operation.

3.1.2 Transformer Alternatives

Beyond attention refinements, emergent paradigms supplant Transformer backbones with equally potent yet more efficient architectures in VLAs, exemplified by Mamba [37]-led state-space models (SSMs) that deliver formidable sequence modeling with linear computational scaling. RoboMamba [19] inaugurates Mamba’s adoption in VLAs as the language backbone, radically obviating Transformer’s quadratic bottlenecks to foster streamlined embodied reasoning. FlowRAM [67] builds upon this foundation by intimately coupling Mamba with conditional flow matching and dynamic radius scheduling, amplifying efficiency and precision in high-precision manipulation scenarios. These alternatives herald a shift toward leaner VLA architecture, synchronously developing architectural thrift with uncompromised multimodal performance.

3.1.3 Efficient Action Decoding

In VLAs, the canonical action decoding paradigm casts images and language instructions as prompts, discretizing continuous actions into binned tokens that the vision-language model autoregressively generates as responses [1], thereby unifying motor outputs within a cohesive token stream. Yet, this autoregressive approach harbors inherent vulnerabilities, chiefly cumulative inference latency from token-by-token generation, which hampers real-time embodied control. To counter these, parallel decoding paradigms—exemplified by Jacobi decoding’s iterative synchronization and speculative decoding’s draft-verification cascades—accelerate autoregressive streams by predicting multiple tokens in parallel, all while curbing fidelity loss. Furthermore, generative modeling-based modules displace autoregressive chains with holistic trajectory synthesis, leveraging latent distributions to evade sequential bottlenecks and deliver deterministic, low-latency action sequences for nimble robotic agents.

Parallel Decoding. Parallel decoding paradigms alleviate autoregressive latencies in VLAs by orchestrating concurrent token predictions, thereby catalyzing real-time embodied actuation. OpenVLA-OFT [14], an extension of OpenVLA [1], pioneers bidirectional attention masks to supplant causal ones, enabling single-pass forward propagation to forecast action chunks of length K in parallel, a tactic echoed in EdgeVLA’s [83] analogous framework for streamlined decoding. PD-VLA [69] reframes autoregressive sequences as nonlinear fixed-point equations solvable via parallel Jacobi iterations, converging in far fewer steps than sequence length n to yield holistic action trajectories with minimal overhead. CEED-VLA [72] refines this by relaxing convergence thresholds for early-exit decoding, mitigating PD-VLA’s stringent criteria-induced inefficiencies, while enforcing output fidelity through consistency distillation to preserve performance. Spec-VLA [75] inaugurates speculative decoding in VLAs, augmented by a relaxed acceptance mechanism that amplifies draft token acceptance rates and mean lengths, precipitating marked inference accelerations.

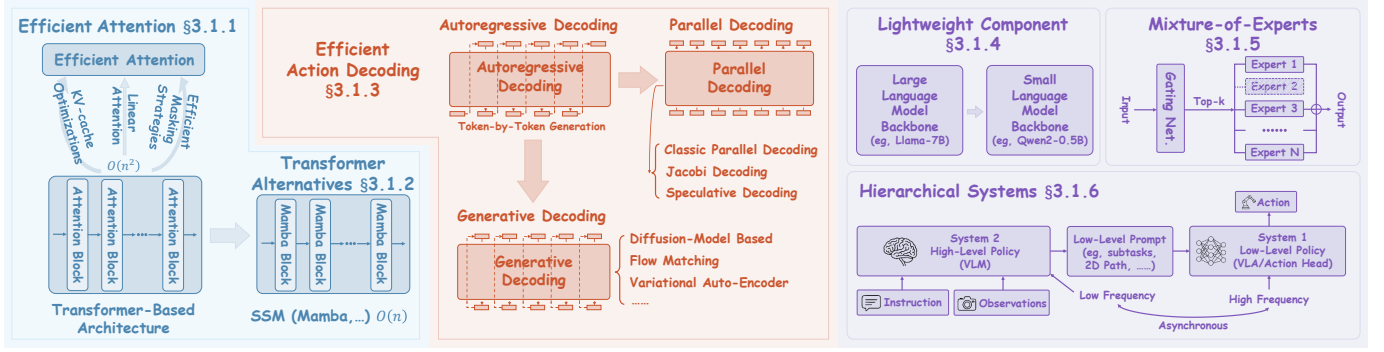


Fig. 5: Key strategies for **Efficient Architectures** (section 3.1) in VLAs. We illustrate six primary approaches: (a) Efficient Attention (section 3.1.1), mitigating the $O(n^2)$ complexity of standard self-attention; (b) Transformer Alternatives (section 3.1.2), such as Mamba; (c) Efficient Action Decoding (section 3.1.3), advancing from autoregressive generation to parallel and generative methods; (d) Lightweight Components (section 3.1.4), adopting smaller model backbones; (e) Mixture-of-Experts (section 3.1.5), employing sparse activation via input routing; and (f) Hierarchical Systems (section 3.1.6), which decouple high-level VLM planning from low-level VLA execution.

Generative Decoding. Pioneering generative paradigms for action sequences in VLAs, TinyVLA [68] inaugurates Diffusion Policy [93] as a dedicated decoder, directly synthesizing continuous robotic actions and circumventing discrete tokenization’s rigidity. HybridVLA [70] elevates this synergy by co-locating diffusion and autoregressive modeling within a singular Transformer scaffold, compressing DDIM [40] sampling to a parsimonious four steps sans efficacy erosion, thus catalyzing inference alacrity. FreqPolicy [71] augments flow-based policies with frequency-domain consistency constraints, exploiting action sequences’ temporal coherence to linearize probability flows and expedite single-step inference while safeguarding sequential integrity. FlowRAM [67] embeds Conditional Flow Matching into robotic policy synthesis, regressing deterministic vector fields to deftly evade diffusion’s iterative denoising cascades, yielding swift inference across minimal timesteps. MinD [73] forges an efficient generative decoding strategy by conditioning its diffusion policy on emergent single-step latents from a predictive world model, eschewing costly full-frame video generation and affirming that compact, information-dense future representations suffice for high-fidelity control signals. VQ-VLA [74] pioneers VQ-VAE [94] for action trajectory discretization, cascading a pretrained VQ decoder atop OpenVLA to distill high-fidelity sequences with unyielding efficiency. AMS [76] refines diffusion decoding through cached latent vectors and vetted success trajectories, judiciously pruning denoising iterations for accelerated convergence. NinA [77] supplants diffusion with normalizing flow decoders, harnessing invertible one-shot sampling to exorcise latency specters inherent in generative chains. Culminating these advances, Discrete Diffusion VLA [78] fuses diffusion’s progressive refinement with a discrete-token interface, enabling adaptive “easy-first, hard-later” decoding and re-masking for error correction—preserving VLM priors, sidestepping autoregressive bottlenecks, and thereby aligning action decoding with VLM transformers to harness unified scaling for expansive VLAs.

3.1.4 Lightweight Component

Lightweight components furnish the most straightforward conduit to efficient VLAs, distilling parametric essence while upholding multimodal prowess and action acuity.

RoboMamba [19] exemplifies this ethos with a mere 3.7M-parameter MLP policy head—constituting just 0.1% of total parameters—to deftly forecast 6-DoF end-effector poses, slashing overhead without curtailing precision. TinyVLA [68] pioneers a compact scaffold, pairing a pre-trained lightweight VLM (<1.4B parameters) on universal vision-language corpora with a diffusion-based policy decoder, yielding unprecedented inference velocity and data thriftiness alongside intact operational fidelity and generalization. Likewise, Edge-VLA [83] and MiniVLA [84] (a lightweight variant of OpenVLA) assemble a 1B-parameter model via Qwen2-0.5B SLM backbone integrated with SigLIP and DINOv2 visual encoders, supporting edge deployment with commendable compactness. CLIP-RT [79] repurposes frozen pretrained CLIP as a unified encoder, paring parameters to one-seventh of OpenVLA’s [1] 7B (down to 1B) yet surpassing its average success rate by 24%, thus inverting scale-performance axioms. Among Diffusion-VLA [95] variants, the minimalist DiVLA-2B leverages Qwen2-VL-2B [96] as VLM backbone, clocking 82 Hz on a solitary A6000 GPU to epitomize throughput supremacy. SVLR [80] innovates by amalgamating sundry lightweight pretrained modules—encompassing Mini-InternVL [97] for vision-language, CLIPSeg [98] for zero-shot segmentation, Phi-3 [99] for language modeling, and all-MiniLM [100] for sentence embeddings—into a retraining-free VLA edifice, enabling scalable task generalization on consumer-grade hardware. NORA [81] harnesses Qwen2.5-VL-3B [101] as core, synergizing with the FAST+ [21] action tokenizer to rival or eclipse far bulkier VLAs in efficacy. SP-VLA [82] unveils an action-aware scheduler that partitions sequences into deliberative and intuitive segments, dynamically invoking heavyweight VLAs or lightweight predictive generators for frequency-adaptive acceleration with negligible performance decrement.

These innovations unlock the promise of ever-lighter

TABLE 2: Representative Works on Efficient Architectures.

Method/Model	Year	Key Innovation in Efficient Architectures
(A) Efficient Attention		
SARA-RT [62]	2024	Introduces up-training to transform quadratic transformers into linear-attention models for efficient VLAs.
Long-VLA [63]	2025	Uses phase-aware input masking to optimize attention focus in long-horizon efficient VLAs.
RetoVLA [1]	2025	Reuses discarded register tokens to enhance spatial reasoning in VLAs.
KV-Efficient VLA [1]	2025	Employs RNN-gated chunked KV cache to accelerate and compress attention in VLAs.
dVLA [1]	2025	Integrates prefix attention masking and KV caching for inference speedup in diffusion VLAs.
(B) Transformer Alternatives		
RoboMamba [19]	2024	Replaces Transformers with Mamba for linear-complexity reasoning in efficient VLAs.
FlowRAM [67]	2025	Integrates region-aware Mamba fusion with flow matching for fast robotic policies.
(C) Efficient Action Decoding		
TinyVLA [68]	2024	Distills large VLAs for fast, data-efficient generative action decoding.
PD-VLA [69]	2025	Reformulates autoregressive decoding into parallel fixed-point iterations for efficient VLA action chunking.
OpenVLA-OFT [14]	2025	Enables parallel action decoding using bidirectional attention in fine-tuned continuous VLAs.
HybridVLA [70]	2025	Merges diffusion and autoregression in unified models for hybrid generative decoding.
FreqPolicy [71]	2025	Uses frequency consistency to optimize flow-based generative action policies.
CEED-VLA [72]	2025	Enhances Jacobi decoding via consistency distillation and early-exit for parallel VLA efficiency.
FlowRAM [67]	2025	Combines Mamba fusion with flow matching for rapid generative manipulation.
MinD [73]	2025	Develops dual-system world models for generative real-time VLA planning.
VQ-VLA [74]	2025	Scales quantized tokenizers for smoother discrete generative action outputs.
Spec-VLA [75]	2025	Introduces speculative decoding with relaxed acceptance for parallel token verification in VLAs.
AMS [76]	2025	Applied OS-level primitives to enhance generative action efficiency in VLAs.
NinA [77]	2025	Trained VLA decoders using normalizing flows for continuous action generation.
Discrete Diffusion VLA [78]	2025	Embeds discrete diffusion for scalable unified action decoding.
(D) Lightweight Component		
RoboMamba [19]	2024	Integrates a lightweight action decoder for parameter-efficient robotic policy fine-tuning
TinyVLA [68]	2024	Distills knowledge from large models into compact VLAs with reduced parameters.
CLIP-RT [79]	2024	Adapts a pre-trained CLIP backbone, achieving 7x fewer parameters for efficient policies.
SVLR [80]	2025	Combines small, pre-trained models in a modular framework for consumer-grade GPU deployment.
NORA [81]	2025	Employs 3B-parameter Qwen backbone with action token compression for size reduction.
SmolVLA [20]	2025	Enables single-GPU training via lightweight architecture for affordable robotics.
SP-VLA [82]	2025	Applies action-aware scheduling with a lightweight action generator to eliminate temporal redundancies.
EdgeVLA [83]	2025	Utilizes small language models, eliminating autoregression for edge device efficiency.
MiniVLA [84]	2025	Reduces parameter count 7x using a smaller LLM while maintaining model performance.
(E) Mixture-of-Experts		
GeRM [85]	2024	Integrates MoE in VLA for faster inference and higher model capacity.
FedVLA [86]	2025	Introduces dual gating MoE for adaptive expert activation in federated VLAs.
TAVP [87]	2025	Employs MoE visual encoder to disentangle features across tasks in VLAs.
(F) Hierarchical System		
HiRT [88]	2024	Implements hierarchical transformers that enable low-frequency VLM to guide high-frequency policy for efficiency.
RoboDual [89]	2024	Synergizes generalist VLA with specialist diffusion for precise multi-step action generation.
DP-VLA [90]	2024	Adopts a dual-process framework with a large reasoning model directing a small sensory executor.
HAMSTER [91]	2025	Deploys high-level VLM producing 2D paths to direct low-level manipulation policies.
Fis [92]	2025	Unifies dual systems by embedding execution in reasoning via parameter sharing.
MinD [73]	2025	Utilizes asynchronous diffusions for low-frequency prediction and high-frequency action coordination.

components in VLAs, collectively catalyzing scalable embodied intelligence.

3.1.5 Mixture-of-Experts

Mixture-of-Experts (MoE) architectures engender efficiency in VLAs by routing tokens to specialized subnetworks, activating only a fraction of parameters to amplify capacity sans commensurate inference costs.

GeRM [85] pioneers sparse MoE integration into quadruped reinforcement learning, demonstrating that selective parameter activation scales model expressiveness for multitask generalization while upholding inference thrift, thereby inaugurating a paradigm for high-fidelity, computationally viable VLA policies. FedVLA [86] advances this

with Dual Gating Mixture-of-Experts (DGMoE) mechanism, transcending conventional top-K routing’s unidirectional token-expert selection via self-aware experts endowed with bidirectional affinities, dynamically sparsifying computational graphs to sustain task efficacy amid resource austerity. TAVP [87] further refines the motif through Task-Aware MoE (TaskMoE), which conditionally engages task-specific experts on amalgamated linguistic-visual signals, disentangling representations across heterogeneous manipulation domains while fostering parameter efficiency via semantically clustered routing gates.

Collectively, these MoE instantiations herald a trajectory toward modular, adaptive VLAs that equilibrate expansive generalization with pragmatic deployment.

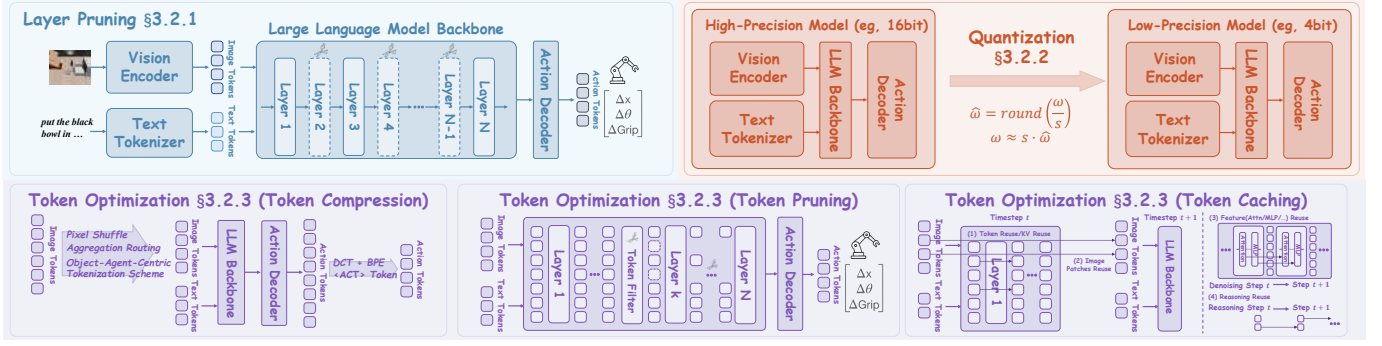


Fig. 6: Key strategies for **Model Compression** (section 3.2) in VLAs. We illustrate three primary approaches: (a) Layer Pruning (section 3.2.1), which removes redundant layers to reduce model depth and computational cost; (b) Quantization (section 3.2.2), which reduces the numerical precision of model parameters to decrease memory footprint and accelerate inference; and (c) Token Optimization (section 3.2.3), which minimizes the number of processed tokens via token compression (merging tokens), token pruning (dropping non-essential tokens), and token caching (reusing static tokens).

3.1.6 Hierarchical Systems

Hierarchical systems in VLAs draw seminal inspiration from psychological dual-process theories, such as [102], [103], bifurcating cognition into deliberate, semantics-rich deliberation and intuitive, rapid execution to emulate human-like embodied agency. These paradigms decouple compute-intensive vision-language model (VLM) inference (System 2) from latency-sensitive action generation (System 1), orchestrating asynchronous execution to harmonize profound semantic comprehension with instantaneous control imperatives.

HiRT [88], DP-VLA [90], and SmolVLA [20] inaugurate such hierarchical scaffolds, wherein languid yet semantically dense VLM representations asynchronously steer lightweight, high-cadence policies, striking an optimal balance between generalization capacity and real-time actuation fidelity. RoboDual [89] operationalizes this duality with OpenVLA as a high-level planner that provides discretized actions and latent encodings, augmented by a lightweight Diffusion Transformer specialist for rapid enactment; latency-aware training therein equips the specialist to redress temporal misalignments from asynchronous interplay, enhancing synergistic robustness. HAMSTER [91] harnesses upper-echelon VLMs to engender executable 2D trajectory sketches that guide specialized low-level policies, melding VLM’s expansive generalization with compact policy thrift. FiS [92] advances inter-system cohesion via partial parameter sharing, embedding the System 1 executor within the VLM-centric System 2 to foster seamless inference-execution orchestration. Fast ECoT [104] extends this asynchronous inference ethos to the Embodied Chain-of-Thought (ECoT) reasoning framework, strategically decoupling latent deliberation from action streams to enable parallel cognitive refinement and output generation, thereby amplifying reasoning depth without temporal encumbrances. MinD [73] consummates this lineage by co-adjudicating a low-frequency generative world model for protracted scene foresight with a high-frequency diffusion policy for on-the-fly control, conditioning the latter on semantically potent single-step latents to enable effective real-time decision-making without exorbitant computational

toll.

Collectively, these resilient hierarchical VLA pipelines scale cognitive depth with operational alacrity.

3.2 Model Compression

As illustrated in Fig. 6, this subsection systematically surveys pivotal techniques, from layer pruning and quantization to token optimization, elucidating their general schemes and how they benefit efficiency to cultivate robust, edge-compatible architectures. Tab. 3 summarizes the representative model compression methods.

3.2.1 Layer Pruning

Layer pruning, the most straightforward paradigm in model compression, precisely excises redundant layers via dynamic mechanisms such as early exits or selective layer skipping, yielding pronounced reductions in parameter counts and inference latency for VLAs. Motivated by the massive interlayer redundancy in large language models—where adjacent layers exhibit high interlayer cosine similarity [18], [22]—this approach unlocks efficiency without eroding the nuanced multimodal synergies essential for embodied tasks. We delineate two principal categories: training-free and training-based strategies.

Training-free methods expedite deployment by leveraging post-hoc analysis. Pioneering this vein, DeeR-VLA [105] introduces a dynamic early-exit framework that circumvents subsequent layers upon ensuring action prediction consistency, thereby slashing computational overhead while preserving competitive task performance. SmolVLA [20], inspired by foundational pruning heuristics, adopts a pragmatic, naive strategy—skipping a fraction $N = 2/L$ of layers—for real-world applicability in resource-scarce settings. RLRC [106] uses Taylor importance criteria to gauge and eliminate low-contribution layers, achieving an overall aggressive sparsity of 90%. FLOWER [108] tailors strategic pruning to architecture: for encoder-decoder VLMs like Florence-2 [122], it discards the entire decoder, retaining only the encoder to halve layer counts; for decoder-only variants like SmolFlow2-Video [123], it prunes the terminal 30% of layers.

TABLE 3: Representative Works on Model Compression.

Method/Model	Year	Key Innovation in Model Compression
(A) Layer Pruning		
DeeR-VLA [105]	2024	Deploys multi-exit architecture for dynamic early termination of redundant layers in VLAs.
MoLe-VLA [18]	2025	Routes spatial-temporal states to selectively skip non-essential LLM layers in VLAs.
SmolVLA [20]	2025	Prunes redundant $N = L/2$ layers in the VLM for compact VLA deployment.
EfficientVLA [22]	2025	Analyzes inter-layer redundancies to prune inconsequential language layers training-free.
RLRC [106]	2025	Applies structured layer pruning followed by RL-based recovery in compressed VLAs.
LightDP [107]	2025	Optimizes pruning recoverability in denoising modules via a unified retraining pipeline.
FLOWER [108]	2025	Fuses intermediate modalities to prune 50% of LLM layers for diffusion head capacity.
(B) Quantization		
OpenVLA [1]	2024	Demonstrates quantization for serving 7B-parameter VLAs without degrading downstream task success rates.
QAIL [109]	2024	Integrates quantization-aware fine-tuning in imitation learning to boost robustness against low-bit precision errors.
FAST [21]	2025	Applies DCT-based tokenization to enable efficient quantization of high-frequency action sequences.
SQIL [110]	2025	Employs saliency-based loss weighting during quantization-aware training to prioritize mission-critical states.
BitVLA [111]	2025	Achieves 1-bit ternary quantization across VLAs with distillation for vision encoder alignment.
RLRC [106]	2025	Employs RL to recover performance after pruning, enhancing robustness to subsequent 4-bit quantization.
SQAP-VLA [112]	2025	Co-designs quantization with token pruning via awareness criteria for incompatibility resolution.
(C) Token Optimization		
FAST [21]	2025	Compresses action tokens effectively via DCT and BPE for enhanced high-frequency VLA efficiency.
VLA-Cache [113]	2025	Caches static visual tokens dynamically with task-aware eviction for improved sequential VLA speedup.
HybridVLA [70]	2025	Caches KV states before diffusion tokens for better iterative denoising efficiency in VLAs.
FlashVLA [114]	2025	Prunes visual tokens precisely via ICS and reuses actions based on stable tokens.
SmolVLA [20]	2025	Compresses visual tokens down to 64 per frame using advanced pixel shuffle.
Fast ECoT [104]	2025	Caches high-level reasoning tokens efficiently for reuse across multiple ECoT timesteps in VLAs.
EfficientVLA [22]	2025	Prunes visual tokens dynamically with task-relevance guidance and caches intermediates for efficient action decoding.
SP-VLA [82]	2025	Prunes spatio-semantic tokens adaptively through velocity-correlated retention ratios for better performance.
CronusVLA [115]	2025	Caches motion features securely in FIFO queues for efficient multi-frame VLA inference.
VOTE [116]	2025	Compresses action chunks into single <ACT> tokens precisely for VLA decoding acceleration.
AMS [76]	2025	Compresses action contexts through hashing-based indexing and evicts low-priority KV caches using LRU-priority.
CogVLA [117]	2025	Aggregates and prunes visual tokens instruction-driven precisely with FiLM routing and percentile thresholds.
SpecPrune-VLA [118]	2025	Prunes tokens dynamically using global-local attention mechanisms and velocity control strategies.
SQAP-VLA [112]	2025	Prunes tokens in a quantized model effectively with robot-aware rings and spatial sampling criteria.
LightVLA [119]	2025	Prunes visual tokens differentially via query-based Gumbel-softmax selection for optimization.
KV-Efficient VLA [66]	2025	Prunes KV chunks selectively with RNN gating for recurrent context summarization efficiency.
ADP [120]	2025	Prunes vision tokens dynamically with trajectory-gated retention and detailed text guidance.
Oat-VLA [121]	2025	Compresses to object-centric and agent-centric tokens via average pooling and gripper-guided selection.

In contrast, training-based schemes infuse adaptability through optimization. MoLe-VLA [18] reconceptualizes LLM layers as distinct experts, deploying a lightweight Spatial-Temporal Aware Router (STAR) to parse visual-spatial and linguistic-temporal cues, dynamically electing and bypassing superfluous layers for tailored execution. LightDP [107] harmonizes learnable layer pruning with consistency distillation via SVD-based importance estimation and the Gumbel-Softmax trick [124], [125], orchestrating a unified framework that dynamically prunes Diffusion Transformer layers during training to amplify compression efficacy.

3.2.2 Quantization

Quantization, a pivotal pillar in model compression, discretizes continuous weights and activations into lower-bit representations, curtailing memory footprints and accelerating inference for VLAs while preserving the fidelity of multimodal action synthesis.

Pioneering empirical validation, OpenVLA [1] systematically probes quantization’s efficacy in large-scale VLAs, demonstrating that aggressive 4-bit post-training quantization halves GPU memory demands while sustaining real-world robotic task proficiency comparable to full-

precision baselines. QAIL [109] advances this frontier with a Quantization-Aware Imitation Learning framework, incorporating a Quantization-Robust Behavior Cloning (QBC) loss that explicitly aligns quantized policy action distributions with their full-precision counterparts, reducing error accumulation in sequential decision-making and facilitating performant edge-device orchestration. SQIL [110] introduces a saliency-aware quantization paradigm, reclaiming near-full-precision performance under 4-bit austerity and yielding up to 2.5× inference speedup in authentic robotic benchmarks. BitVLA [111] innovates with a distillation-aware training paradigm to pioneer 1-bit quantization in VLAs, seamlessly embedding LLM backbones and visual encoders into ternary parameter spaces $\{-1, 0, 1\}$, vindicating extreme post-training quantization’s viability with 3.36× memory compression and competitive prowess in intricate manipulation tasks. RLRC [106] augments this toolkit via a performance recovery pipeline, strategically deploying reinforcement learning fine-tuning to empower subsequent 4-bit quantization with maximal memory thrift without performance degradation. Extending quantization to ancillary efficiencies, FAST [21] repurposes it for token compression, forging a systematic action quantization scaffold that transmutes continuous action into information-

rich discrete latent spaces via spectral decomposition. SQAP-VLA [112] integrates quantization and token pruning via Pruning-Targeted Quantizer Enhancement, applying Hadamard transforms to the weights and activations of the query and key layers prior to low-bit discretization, thereby mitigating quantization-induced distortions in attention maps to foster their interpretability and resilience for salient token selection.

Collectively, these advancements underscore quantization’s transformative role in sculpting VLAs for ubiquitous, resource-frugal embodiment.

3.2.3 Token Optimization

Token optimization, an elegant facet of model compression, strategically refines the sequence of representational tokens in VLAs, thereby mitigating the quadratic escalation of transformer-based computations in inference while safeguarding the performance of integrated visual-linguistic-actional reasoning. Through targeted mechanisms like compression, pruning, and caching, this paradigm curtails token redundancy, alleviating memory pressures and expediting the temporal dynamics of decision-making in embodied systems, thus enabling agile deployment on resource-limited robotic platforms. This subsection methodically reviews representative methods, illuminating their seamless integration with decoder frameworks to yield lean, high-fidelity multimodal inference pipelines.

Token Compression. Token compression distills voluminous token streams into succinct representations by algorithmically aggregating informational essence, thereby curtailing the computational swell in VLAs and bolstering inference velocity without forfeiting multimodal fidelity.

For vision tokens, SmolVLA [20] enforces spatial thrift via pixel shuffle operations, confining visual tokens to a mere 64 per frame to temper the deluge of perceptual inputs. CogVLA [117] advances aggregation routing with Encoder-FiLM modules [126], coalescing myriad patch tokens into sparse, instruction-driven aggregates that yield substantial computational economy, while preserving—or even augmenting—cross-modal coherence for action synthesis. Oat-VLA [121], on the other hand, pioneers an object-agent-centric tokenization scheme that strategically injects structural inductive biases into visual processing, resulting in an order-of-magnitude compression of vision tokens.

Shifting to action tokens, FAST [21] pioneers spectral-domain parsimony: it applies discrete cosine transform (DCT) to normalized action sequences, transmuted signals into frequency components, then refines them via byte-pair encoding (BPE [127]) into compact, information-dense token cascades that streamline sequential policy generation. VOTE [116], in turn, pioneers extreme condensation by emitting a singular <ACT> token to encapsulate prospective action trajectories, decoded post hoc through a lightweight MLP action head into precise continuous maneuvers—thus dramatically contracting the output streams.

Token Pruning. Token pruning algorithmically excises redundant tokens to preserve only indispensable ones, thereby streamlining computational demands in VLAs and enhancing inference efficiency without impairing multimodal coherence. This technique harnesses salience metrics

and adaptive heuristics to distill token sequences, fostering agile embodied reasoning amid resource constraints.

Inspired by FastV [128], which is a classic representative of token pruning schemes in the field of VLMs, FlashVLA [114] proposes an information contribution score (ICS)-guided pruning mechanism, devising a training-free, Flash Attention [129]-compatible framework that sets a plug-and-play paradigm for VLA inference acceleration. Building thereon, EfficientVLA [22] deploys a multi-step Task-Relevance and Diversity-Driven Visual Token Pruning scheme, harmonizing task saliency with feature heterogeneity to forge a concise, richly informative token ensemble. SP-VLA [82] refines spatial-semantic perceptual capability through Dual-Aware Token Pruning, jointly evaluating semantic and positional importance while dynamically tuning pruning ratios guided by velocity scales for nuanced adaptation. Advancing cognitive integration, CogVLA [117] embeds instruction sensitivity within the LLM backbone’s pruning pipeline, engendering lean representations that amplify semantic potency. SpecPrune-VLA [118] experimentally points out that EfficientVLA’s [22] strategy of relying only on local information for pruning is unreliable and pioneers self-speculative pruning by exploiting temporal continuity from prior inferences to inform current token curation, yielding marked speedups in VLA dynamics. SQAP-VLA [112] fortifies resilience with Quantization-Aware Token Pruning, adeptly pinpointing vital tokens despite quantization’s skewing of statistical distributions in features such as attention scores, and ultimately outperformed the EfficientVLA [22] in experiments. Culminating this progression, LightVLA [119] introduces an adaptive, performance-oriented visual token pruning framework, generating dynamic queries for importance gauging and Gumbel softmax for seamless, differentiable token selection, solving the problem of EfficientVLA’s [22] reliance on fixed pruning ratios. Furthermore, KV-Efficient VLA [66] introduces a two-stage token pruning mechanism, which first splits the historical KV Cache into fixed-size chunks, aggregates them into a single compressed representation, and then applies a lightweight RNN to prune these chunks through a threshold.

Like SP-VLA [82] and SpecPrune-VLA [118], ADP [120] introduces an action-aware gating token pruning mechanism. The difference is that SP-VLA [82] and SpecPrune-VLA [118] set the token pruning ratio based on the motion speed of the end effector, while ADP [120] dynamically decides whether to prune or not based on the displacement of the end effector trajectory within a small period of time. In addition, ADP introduces a text-driven pruning mechanism, which is able to compute the similarity between vision tokens and task instructions based on cross-modal attention and retains the most relevant Top-K tokens, enriching the semantic information of token pruning.

Token Caching. Token caching, the capstone of token optimization, stockpiles reusable tokens for iterative reuse in subsequent processing stages, thereby eliminating redundant computations in VLAs and enhancing inference efficiency while maintaining multimodal temporal coherence. This strategy exploits temporal invariances across frames or steps, transforming static or stable representations into persistent assets that undergird fluid embodied trajectories.

VLA-Cache [113] forges reusable token repertoires by pinpointing static tokens with minimal inter-frame variance and evicting task-relevant ones, leveraging KV-cache for seamless redeployment in ensuing iterations. Hybrid-VLA [70] ingeniously extrapolates KV-caching—the bedrock of autoregressive decoding—to diffusion models’ iterative denoising, hoarding invariant conditional token key-value pairs to excise inter-step redundancies, thus amplifying inference velocity without compromising diffusion-driven continuous action fidelity. FlashVLA [114] introduces a token-aware action reuse protocol, judiciously recycling prior actions via metrics of action and token stability to harness sequential consistencies. Fast ECoT [104] infuses caching into Embodied Chain-of-Thought (ECoT) reasoning, reusing swaths of inference chains rather than regenerating them, compacting computational graphs for precipitous latency reductions. EfficientVLA [22] critically points out that the VLA-Cache’s [113] approach is limited by LLM’s memory bottleneck and deploys static caching of intermediate self-attention and MLP features across denoising steps, sidestepping iterative redundancies in generation loops. CronusVLA [115] pioneers feature-level token caching via a FIFO queue that stockpiles and recycles compact motion features, decoupling compute-intensive single-frame perception from lightweight multi-frame inference. AMS [76] sustains a GPU-resident Context Pool archiving prior inference intermediates, inaugurating a hardware-aware, holistic caching paradigm that transcends conventional key-value bounds to encompass latent vectors and output embeddings across the whole VLA pipeline.

In essence, token caching increases VLA efficiency by perpetuating representational continuity across temporal horizons.

3.3 Discussion

3.3.1 Innovations

Efficient VLA design has transitioned from scaling-centric paradigms to adaptive architectures that integrate linear-time attention and hierarchical decoupling to alleviate inherent computational bottlenecks. This evolution, bolstered by compression strategies such as pruning and quantization, prioritizes the retention of perceptual-motor invariants through task-aware distillation. Such advancements effectively reconcile high-level reasoning with reactive execution, enabling the deployment of foundational intelligence within the stringent resource constraints of physical agents.

3.3.2 Limitations

Nevertheless, aggressive structural optimization often engenders a tension between architectural parsimony and representational fidelity, potentially inducing semantic drift that undermines dexterity in long-horizon tasks. Hierarchical and parallelized frameworks also struggle with spatiotemporal coherence, where asynchronous execution leads to unstable control signals. Furthermore, the reliance on static importance metrics limits environmental adaptability, necessitating future research into hardware-algorithm co-design to optimize the Pareto frontier of efficiency and robustness.

4 EFFICIENT TRAINING

While foundational VLAs leverage pre-trained VLM backbones for robust multimodal reasoning, they inherit prohibitive computational and data overheads, necessitating the advancement of efficient training methodologies to alleviate resource demands without sacrificing model performance. As illustrated in Fig. 7, this chapter systematically explores the spectrum of these techniques, focusing on two pivotal stages: (1) **Efficient Pre-training**, instilling foundational action capabilities into pre-trained VLMs or training the entire VLA from scratch with minimal overhead, and (2) **Efficient Post-training**, enabling the swift and effective deployment of VLAs to specific downstream tasks.

A VLA-centric Taxonomy of Training. Our taxonomy is defined from a functional, VLA-centric perspective. We define **Pre-training** as the entire process of migrating a general-purpose VLM into the embodied domain to create an initial, action-aware policy. This foundational stage is concerned with instilling the fundamental capability to act. In contrast, **Post-training** focuses on the subsequent specialization of this generalist policy, adapting it to excel in specific tasks, environments, or under particular resource constraints. This perspective clarifies why our pre-training VLA stage encompasses a broad spectrum of methods, including some techniques often labeled as “post-training” in other fields [130], [131], [132], [133]. The key distinction is not the method itself, but its objective: *if the goal is to create the first version of the action-capable model from a VLM, we classify it as pre-training*. This approach offers a more fundamental and logically consistent framework that aligns with the practical development lifecycle of VLAs.

4.1 Efficient Pre-Training

As illustrated in Fig. 7 left, this section systematically examines advancements categorized into data-efficient pre-training, efficient action representation, and other efficient pre-training strategies. By highlighting these innovations, we outline the critical pathways for developing robust, general VLA policies without incurring the traditional burden of extensive, resource-intensive pre-training cycles. We list representative works in Tab. 4.

4.1.1 Data-Efficient Pre-training

For the data scarcity problem faced in training VLA models, some works address it by efficient data collection, which will be presented in section 5, and other works focus on data-efficient pre-training. Data-efficient pre-training addresses the VLA data dependency by prioritizing the judicious use of both scarce robotic trajectories and readily available large-scale non-robotic data, such as human ego-centric videos, to overcome the prohibitive cost and redundancy of vast multimodal corpora. This subdomain bifurcates into two principal strategies: leveraging unlabeled data via sophisticated self-supervised training objectives and bridging domain gaps through mixed data co-training frameworks.

Self-Supervised Training. Self-supervised training represents a crucial paradigm for enhancing data efficiency in VLA models by synthesizing effective supervisory signals from unlabeled or readily available datasets, thereby mitigating the substantial data scarcity inherent in embodied learning. This methodology bifurcates into two main

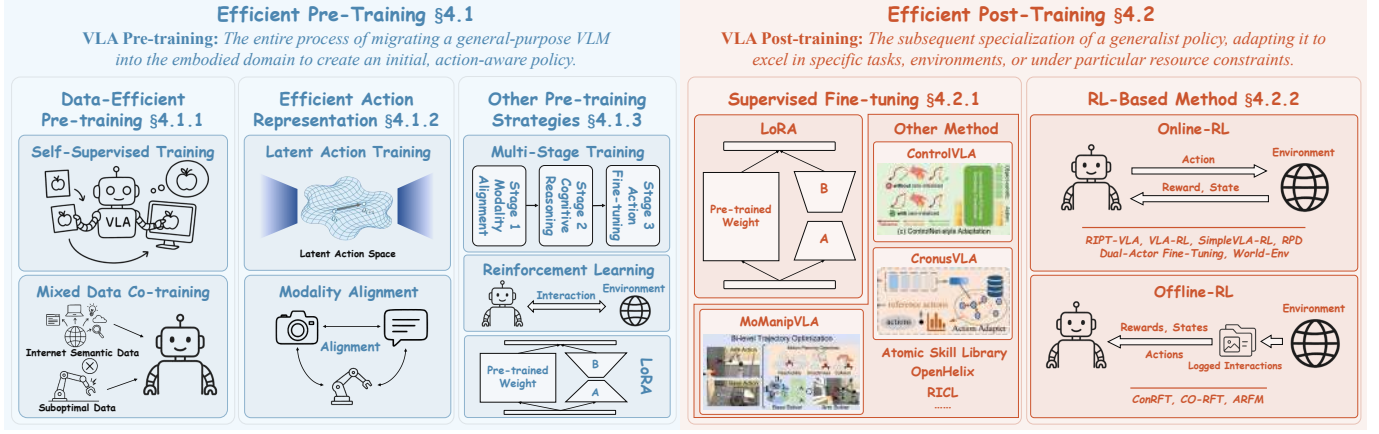


Fig. 7: Key strategies for **Efficient Training** (section 4) in VLAs, divided into two main stages. (a) Efficient Pre-Training (section 4.1) migrates general-purpose VLMs into the embodied domain to create an initial, action-aware policy, encompassing Data-Efficient Pre-training (section 4.1.1), Efficient Action Representation (section 4.1.2), and Other Pre-training Strategies (section 4.1.3). (b) Efficient Post-Training (section 4.2) subsequently specializes this policy for specific tasks, leveraging Supervised Fine-tuning (section 4.2.1) and RL-Based Methods (section 4.2.2).

strategies: (1) augmenting limited expert trajectories and (2) harnessing internet-scale ego-centric video.

The first approach focuses on maximizing the utility of existing robotic data. DTP [134] strategically employs a diffusion-based trajectory prediction model as an efficient and scalable pre-training objective. By learning the generated future end-effector trajectories in the RGB domain, DTP effectively bridges the modality gap between high-level visual-language input and the continuous physical action space, consequently improving sample efficiency and long-horizon generalization in imitation learning.

The second, and highly prevalent, strategy centers on utilizing massive, unlabeled internet-scale ego-centric videos to alleviate the severe data dependency of VLA models. Early works, such as LAPA [135], innovated by entirely circumventing the need for expensive real robot action labels. It achieves this by learning a discrete latent action space from vast amounts of unlabeled video data. However, Bu et al. [136] critically noted that LAPA’s [135] raw pixel-based reconstruction objective inadvertently encodes task-irrelevant dynamics (e.g., camera shake, background motion), which severely interferes with subsequent policy training. To address this, they proposed Task-centric Latent Action Learning, which utilizes a two-stage VQ-VAE to decouple and quantize purely task-centric latent actions from the video stream, further increasing data efficiency. LAWM [137] later extended LAPA’s [135] latent action learning principle, applying it to more compact architectures like BAKU [138] and RSSM [139]. EgoVLA [140] abandoned latent action spaces and introduced a shared action space based on MANO parameters, effectively translating human self-centric video data into actionable VLA model training. For dexterous manipulation, Being-H0 [141] introduced Physical Instruction Tuning by leveraging part-level motion tokenization (discretizing MANO parameters) on the large-scale UniHand dataset, allowing the VLA model to acquire high-fidelity dexterous manipulation priors with superior sample efficiency. Finally, methods focused on explicit dynamics learning, such as RynnVLA-001’s [142] three-stage

generative pretraining (I2V prediction of future frames) and Wang et al.’s [143] World Model objective, demonstrate how learning environment dynamics from action-less videos can form a robust foundation, which is then fine-tuned with action tokens to acquire the final policy.

Mixed Data Co-training. Mixed Data Co-training is a potent strategy for boosting VLA model efficiency and generalization by jointly training on heterogeneous datasets across varying quality and modality. GerM [85] establishes a generalist robotic model using a Conservative Q-Learning (CQL) offline RL framework, which robustly leverages both expert and suboptimal data by strategically mitigating policy overestimation on out-of-distribution actions. HAM-STER [91] employs a hierarchical VLA structure: a high-level VLM predicts a coarse 2D end-effector trajectory, which conditions a low-level 3D control policy. This decoupling efficiently assimilates massive cross-domain data. Similarly, GraspVLA [50] unifies autoregressive perception and flow-matching action generation within a Chain-of-Thought (CoT) objective, enabling seamless joint training across synthetic and internet semantic data. Furthermore, AnyPos [146] addresses task-specific data dependency by introducing Arm-Decoupled Estimation and a Direction-Aware Decoder. This mechanism extracts generalizable motion primitives from large-scale task-agnostic datasets, substantially enhancing zero-shot transferability.

4.1.2 Efficient Action Representation

Action, as the distinct embodied modality, is frequently high-dimensional, continuous, and noisy, significantly impeding VLA training efficiency and generalization. To address this, several works explore more compact and semantic action representations. This effort can be categorized into action space compression and innovative action modeling.

A primary direction employs compression to transform continuous, high-dimensional actions into a more succinct representation space. This reduces the search space for policy learning. LAPA [144], Bu et al. [136], RynnVLA-001 [142], and LAWM [137] all utilize the autoencoder [150]

TABLE 4: Representative Works on Efficient Pre-training.

Method/Model	Year	Key Innovation in Efficient Pre-training
(A) Data-Efficient Pre-training		
GeRM [85]	2024	Uses mixture-of-experts architecture and offline reinforcement learning to optimize data utilization.
LAPA [144]	2024	Through unsupervised action quantization, latent skills can be extracted from unlabeled videos.
HAMSTER [91]	2025	Hierarchical action modeling minimizes data redundancy for open-world manipulation pretraining.
DTP [134]	2025	Diffusion-based synthesis optimizes trajectory data usage for long-horizon task pretraining.
Humanoid-VLA [145]	2025	Egocentric visual priors reduce data requirements for universal humanoid VLA pretraining.
GraspVLA [50]	2025	Synthetic data scaling enables robust grasping pretraining with minimal real-world samples.
UniVLA (Bu et al.) [136]	2025	Cross-embodiment analysis extracts task-centric latents from sparse videos for adaptive pretraining.
UniVLA (Wang et al.) [143]	2025	Multimodal tokenization unifies sparse data streams for efficient VLA pretraining.
EgoVLA [140]	2025	Transfers expertise from egocentric human videos with minimal data through a unified action space.
AnyPos [146]	2025	Enables training from task-agnostic data for bimanual pretraining by an inverse dynamics model.
Being-H0 [141]	2025	Hand motion priors from human videos minimize dexterous pretraining data via physical instruction tuning.
RynnVLA-001 [142]	2025	Human demonstrations encoded via generative models streamline robot manipulation pretraining.
LAWM [137]	2025	Self-supervised world models forecast latent dynamics, minimizing imitation pretraining data.
(B) Efficient Action Representation		
LAPA [144]	2024	VQ-VAE quantizes unsupervised latent actions from videos for label-free VLA pretraining.
FAST [21]	2025	Uses DCT, BPE, and quantization to compress action sequences, reducing VLA pre-training time.
UniVLA (Bu et al.) [136]	2025	Language-conditioned decoupling derives latent actions from cross-embodiment videos for compact pretraining.
cVLA [147]	2025	Discrete keypose prediction in image space enables lightweight simulation-based VLA pretraining.
EgoVLA [140]	2025	Proposes a shared action space based on MANO parameters and retargets robot demo to human action.
VLA-Adapter [148]	2025	Bridge Attention injects optimal vision-language features into actions for tiny-scale model pretraining.
RynnVLA-001 [142]	2025	ActionVAE compresses chunks into continuous latents for multi-stage pretraining transfer.
ReSET [149]	2025	Restructures initial states with human-derived policies, and significantly reduces reliance on expert presentations.
LAWM [137]	2025	Extract latent actions via joint imitation-world modeling from unlabeled videos.
(C) Other Pre-training Strategies		
RoboMamba [19]	2024	Frozen CLIP integrates with Mamba, followed by lightweight projectors, enabling compute-efficient pre-training.
TinyVLA [68]	2024	Compact VLMs initialize VLAs via LoRA, bypassing extensive robotic datasets for data-efficient training.
TAVP [87]	2025	Accelerated an efficient strategy by a novel pseudo-environment to proactively capture informational perspectives.

principle to distill actions into a latent space. Specifically, LAPA and Bu et al. leverage VQ-VAE [94] to capitalize on both quantization and strong compression capabilities. In contrast, RynnVLA-001 proposes ActionVAE on a standard VAE [151] architecture, while LAWM derives latent action representation from the powerful DreamerV3 [139] world model. Beyond AE, FAST [21] achieves up to a 5× reduction in pre-training time by directly compressing action sequences via algorithms like Discrete Cosine Transform (DCT) and Byte-Pair Encoding (BPE), effectively achieving dimensionality reduction in the sequence domain.

Another set of strategies innovates on how actions are modeled relative to other modalities and defined. Both EgoVLA [140] and VLA-Adapter [148] employ a cross-modal feature linkage strategy to enhance training efficiency. Specifically, EgoVLA bridges human demonstrations to robot action representations through a shared action space based on MANO parameters, while VLA-Adapter utilizes a Bridge Attention module to connect vision-language representations to actions. Furthermore, optimizing the co-ordinate system choice can drastically lower complexity. cVLA [147] eschews the traditional robot base frame, instead representing actions in the image coordinate system before mapping them to the end-effector pose space, facilitating simpler, lower-dimensional action encoding. Finally, ReSET [149] focuses on compressing the dense robotic action state distribution into a manageable set of anchor states, granting policies enhanced generalization capabilities in data-scarce settings.

In essence, efficient action representation shifts VLA

learning from modeling noisy, high-dimensional control to mastering compact, semantic, and transferable motion primitives.

4.1.3 Other Pre-training Strategies

Beyond data-efficient and action-representation techniques, several innovative strategies further enhance VLA pre-training efficiency by leveraging specialized paradigms. Multi-stage training emerges as a prominent approach, adopted in works such as [19], [50], [134], [142], [145], [152]. This method decomposes the training pipeline into sequential phases, decoupling modality alignment, cognitive reasoning, and action fine-tuning. By isolating these components, models acquire sophisticated inference capabilities and robust real-world interaction skills with reduced computational overhead, enabling progressive knowledge distillation across stages.

Reinforcement learning (RL) offers another avenue for efficient VLA pre-training, as demonstrated in [85], [87]. Specifically, [85] employs Conservative Q-Learning to maximize data utilization, mitigating sample inefficiency in exploratory settings. Meanwhile, [87] introduces pseudo-environments to simulate interactions, accelerating convergence by bypassing costly real-world data collection and trial-and-error loops.

Additionally, low-rank adaptation (LoRA [130]) techniques facilitate resource-aware pre-training in [68], [88]. These methods inject lightweight adapters into the VLM backbone, preserving its potent visual perception and semantic comprehension while enabling targeted updates for

TABLE 5: Representative Works on Efficient Post-training.

Method/Model	Year	Key Innovation in Efficient Post-training
(A) Supervised Fine-tuning		
OpenVLA [1]	2024	Explores the pros and cons systematically of different parameter-efficient fine-tuning paradigms.
Atomic Skill Library [153]	2025	Few-shot supervised fine-tuning adapts pre-trained VLAs to atomic skills with minimal data.
OpenVLA-OFT [14]	2025	Parallel decoding and action chunking accelerate LoRA fine-tuning of VLAs for high-frequency bimanual tasks.
MoManipVLA [154]	2025	Adapts pre-trained VLAs to mobile manipulation efficiently using waypoint-guided supervised fine-tuning.
OpenHelix [155]	2025	Refines only the <ACT> token during fine-tuning to boost efficiency with minimal parameter updates.
ControlVLA [156]	2025	Injects object-centric representations into pretrained VLAs with ControlNet-style efficient adaptations.
CronusVLA [115]	2025	Finetunes single-frame pre-trained VLAs into multi-frame decoding using compact historical frame aggregation.
InstructVLA [157]	2025	Enhance pre-trained VLAs from understanding to manipulation with LoRA and MoE-adaptation.
RICL [158]	2025	Inject in-context adaptability into pre-trained VLAs with distance-weighted action interpolation.
ATE [159]	2025	Align and steer pre-trained VLAs for task adaptation using unified latent guidance through latent space fine-tuning.
(B) RL-Based Method		
ConRFT [160]	2025	Unifies consistency-based RL objective for offline policy extraction from few demonstrations.
RPD [161]	2025	Distills VLAs into compact RL experts using MSE-guided on-policy exploration for enhanced sample efficiency.
RIPT-VLA [162]	2025	Empowers RL post-training with dynamic rollout sampling and leave-one-out advantage estimation.
VLA-RL [163]	2025	Applies trajectory-level RL to auto-regressive VLAs using dense rewards from pseudo-labeled task segments.
CO-RFT [164]	2025	Extends TD learning to action chunks in offline RL for VLA fine-tuning.
ARFM [165]	2025	Introduces scaling factor in flow matching loss to balance RL advantage preservation and gradient variance.
SimpleVLA-RL [13]	2025	Enhance scalable RL with VLA-specific trajectory sampling and parallelization under data scarcity.
Dual-Actor Fine-Tuning [166]	2025	Deploy dual actors for multi-task actions and latent refinements guided by human language interventions.
World-Env [167]	2025	Leverage video-based world model as resettable virtual environment for safe RL exploration with minimal demos.

action integration. This modular adaptation minimizes parameter updates, yielding efficient VLAs without compromising foundational multimodal prowess.

4.2 Efficient Post-Training

As illustrated in Fig. 7 right, these methods encompass supervised fine-tuning for targeted parameter updates and reinforcement learning-based approaches for policy optimization. Notably, efficient post-training is the crucial adaptation phase that ensures VLAs are not only specialized but also computationally viable for robust deployment in a wide array of practical applications. Tab. 5 summarize representative works.

4.2.1 Supervised Fine-tuning

Supervised fine-tuning refines pre-trained VLAs by further training on labeled datasets for specific downstream tasks, thereby enhancing task-specific capability while preserving multimodal generalization.

In parameter-efficient fine-tuning, OpenVLA [1] pioneers a systematic exploration of five strategies—full fine-tuning, last-layer-only, frozen vision, sandwich fine-tuning, and LoRA—demonstrating that LoRA fine-tuning strikes an optimal performance-compute trade-off. Subsequent work, OpenVLA-OFT [14], advances this paradigm with an integrated approach combining parallel decoding, action chunking, continuous action representation, and a simple L1-regression-based learning objective, significantly boosting efficiency for edge deployment of VLAs. InstructVLA [157] further innovates by merging LoRA adapters with scaled MoE-adaptation heads, achieving robust parameter-efficient tuning.

Beyond these, the Atomic Skill Library [153] constructs a dynamic, reusable repository of atomic skills via a data-driven three-wheeled approach, enabling low-data-cost post-training adaptation and potent compositional generalization. MoManipVLA [154] introduces a novel bi-level

trajectory optimization framework, leveraging merely 50 real-world samples to seamlessly transfer pre-trained VLAs to mobile manipulation tasks. OpenHelix [155] augments MLLM inputs with a learnable <ACT> token, freezing all MLLM parameters while training only the <ACT> token embedding for cost-effective task adaptation. ControlVLA [156] fuses pre-trained VLAs with object-centric representations via a ControlNet-style architecture, employing zero-initialized projection layers to enable efficient fine-tuning with 10-20 demonstration samples while retaining prior knowledge. CronusVLA [115] enhances single-frame VLAs with multi-frame capabilities by freezing the backbone’s historical frame perception and appending a lightweight cross-frame decoder through minimum computational overhead. RICL [158] integrates In-Context Learning (ICL [168]) into VLA post-training, mimicking RAG processes by concatenating retrieved (image, state, action) sequences with query sequences, training the model to predict actions from contextual cues for few-shot tuning. Lastly, ATE [159] employs reverse KL divergence for structured alignment in latent space, guided by energy-model-defined gradients to steer the sampling of diffusion or flow-matching VLAs toward target distributions with remarkable efficiency.

4.2.2 RL-Based Method

Although supervised fine-tuning excels in leveraging high-quality task-specific data, its efficacy hinges precariously on data abundance and quality, rendering reinforcement learning (RL)-based post-training a potent antidote to scarcity and variability in robotic datasets. These methods bifurcate into online paradigms, which harness real-time environmental interactions to iteratively refine policies, and offline paradigms, which distill expertise from static trajectories without further data acquisition.

Online RL fosters adaptive exploration, as exemplified by RIPT-VLA [162], which integrates sparse binary rewards

with a rejection-sampled PPO variant to elevate success rates from 4% (SFT) to 97% in mere 15 iterations using a single demonstration. Also based on the PPO [169] algorithm, VLA-RL [163] reframes trajectories as multi-turn conversations, deploying PPO with VLM-derived dense rewards and curriculum optimizations to yield 4.5% success gains over OpenVLA (SFT) on 4 LIBERO task suites. SimpleVLA-RL [13] extends this efficiency via GRPO [170] and interactive sampling on OpenVLA-OFT, reducing data needs by boosting rates from 17.3% to 91.7% with one trajectory per task while enhancing sim-to-real transfer. Complementarily, RPD [161] distills teacher VLAs into compact policies using MSE-aligned PPO, surpassing originals in sparse-reward ManiSkill3 [171] tasks with accelerated convergence. A human-in-the-loop dual-actor framework [166] further refines diffusion-based VLAs with a “Talk-and-Tweak” scheme through latent tweaks from language-mapped corrections, attaining 100% success in 101 minutes across tasks with 2× multi-robot efficiency. World-Env [167] innovates by simulating futures in video-based virtual environments with VLM-guided rewards, achieving 79.6% LIBERO success from five demonstrations post-training without real-world costs.

In contrast, offline RL maximizes the utility of archived data. Both ConRFT [160] and CO-RFT [164] use Cal-QL [172], which effectively mitigates the value overestimation problem in offline RL by penalizing the Q-value of out-of-distribution actions and compensating for in-dataset actions. Specifically, ConRFT’s [160] initial Q-learning with consistency objectives initializes stable policies from 20–30 demos, enabling 96.3% real-world success post-online handover with 144% baseline gains, while CO-RFT [164] chunks actions for transformer-critic Q-prediction, lifting success by 57% and trimming cycles 22.3% on 30–60 samples. Furthermore, ARFM [165] caps this lineage with adaptive scaling in flow-matching losses to curb variance, delivering 4.5% multi-task uplift and 11.4% perturbation robustness over π_0 [2] baseline.

Collectively, these RL pipelines empower VLAs with resilient, data-thrifty adaptation, transcending supervised limits toward autonomous prowess.

4.3 Discussion

4.3.1 Innovations

VLA training has shifted toward modular, data-thrifty regimes that facilitate the seamless migration of multi-modal priors into the embodied domain. By leveraging unlabeled egocentric videos, current methodologies instill foundational motor priors while circumventing the prohibitive costs of expert demonstrations. This process is further refined by latent action modeling and spectral compression, which distill high-dimensional control into compact manifolds to streamline the policy search space. Ultimately, decoupling foundational pre-training from task-specific refinement—particularly through reinforcement learning—enables models to achieve fine-grained dexterity without eroding the reasoning depth of their vision-language origins.

4.3.2 Limitations

Nevertheless, the kinematic discrepancy between human-centric video and robotic platforms introduces significant noise, hindering the acquisition of high-fidelity action priors. While parameter-efficient fine-tuning curtails computational overhead, it may constrain the representational flexibility required for the deep cognitive shifts essential in multi-stage manipulation. Furthermore, RL-based adaptation remains susceptible to reward misspecification and distribution shifts, where offline methods suffer from value overestimation and online paradigms face severe sample-efficiency bottlenecks. Addressing these challenges necessitates more robust alignment objectives that reconcile rapid task specialization with the preservation of multimodal coherence in unscripted environments.

5 EFFICIENT DATA COLLECTION

The performance of VLAs depends critically on the scale, quality, and diversity of demonstration datasets across embodiments and task variations. Unlike LLMs and VLMs, which benefit from Internet-scale training data, VLAs cannot directly leverage such resources. Their prevailing data collection paradigm—human teleoperation and expert demonstrations in real-world settings—is inherently labor-intensive, prohibitively costly, and severely limited in scalability. As illustrated in Fig. 8, Recent efforts to overcome these challenges have pursued several strategies, including human-in-the-loop data collection (section 5.1), simulation data collection (section 5.2), Internet-scale and cross-domain data utilization (section 5.3), self-exploration data collection (section 5.4), and data augmentation (section 5.5). We systematically review these strategies, analyzing their core principles and representative methods, with a concise summary provided in Tab. 6.

5.1 Human-in-the-Loop Data Collection

Traditional human-in-the-loop data collection proves costly, labor-intensive, time-consuming, and fundamentally inefficient. Firstly, this approach exhibits strong dependence on expert human operators, specialized hardware, manual annotations, and physical robot deployments in real-world settings with carefully designed scenarios, requiring substantial financial and resource investments. Moreover, the 1:1 ratio between human demonstration time and collected data, further degraded by environment setup, task resets, and human errors, results in efficiency far below expectations. These compounding limitations lead to severe data scarcity, impeding efforts to scale dataset sizes and thereby restricting the generalizability and robustness of VLAs.

To acquire novel robotic data in a faster, more cost-effective, and scalable manner, the most direct approach is to optimize the role and efficiency of human involvement in the data collection loop. Along this approach, recent research has explored methods where humans are repositioned as supervisors or high-level instruction providers who collect data through more efficient interfaces or intervene only at critical junctures.

CLIP-RT [79] collects robot demonstrations via natural language interfaces. Users conversationally interact with an



Fig. 8: **Taxonomy of Efficient Data Collection Strategies in VLAs.** This figure illustrates the primary approaches under section 5, encompassing human-in-the-loop, simulated, reusability-oriented, self-driven, and augmentative techniques for scalable acquisition of high-quality robotic datasets while minimizing resource overhead.

LLM, which translates linguistic commands into low-level end-effector actions. The camera captures observations, the robot executes actions, and the system records complete trajectories. While CLIP-RT eliminates the need for expert knowledge, it still requires continuous human engagement throughout data collection. In contrast, GCENT [173] addresses the constraint by positioning the human operator as a guardian who intervenes only when a failed or near-failure step is detected. Interventions are executed through an interactive rewind and correction mechanism, which allows operators to restore the robot to a previous state and provide corrective demonstrations. And by selectively soliciting human corrections and iteratively refining the policy online, GCENT progressively reduces intervention frequency and increases task success, ultimately enabling a one-operator-multiple-robots setup.

5.2 Simulation Data Collection

While these human-in-the-loop optimization strategies substantially improve efficiency over traditional teleoperation methods, the data achievable through such approaches remains insufficient and monolithic. Alternatively, simulation environments allow data generation to be scaled across a wide range of settings, object types, lighting conditions, and agent embodiments through automated, parallel processes, thus enabling the curation of large-scale datasets with greater diversity at a fraction of the time and expense.

GraspVLA [50] introduces SynGrasp-1B, a billion-frame synthetic grasping dataset generated through photorealistic simulation in parallel. They employ a modular expert policy to produce collision-free trajectories autonomously, validated in MuJoCo and rendered via Isaac Sim. GeRM [85] introduces QUARD-Auto, a training dataset for quadruped manipulation, also generated through massively parallel simulation in Isaac Gym. A pre-trained policy is adopted

to eliminate the need for manual teleoperation. cVLA [147] presents models trained on datasets generated via the ManiSkill simulator. The data generation pipeline employs analytical grasp models and privileged pose information to produce verified action trajectories. RoboTwin 2.0 [16] generated bimanual manipulation data through an expert-simulated pipeline. The automated pipeline employs a closed-loop architecture where a code-generation agent synthesizes task programs from natural language instructions, executes them in simulation, and iteratively refines them based on dual feedback until achieving target success rates. To ensure task feasibility across heterogeneous dual-arm platforms, it also annotates objects with diverse candidate grasp poses and applies robot-specific motion planning. However, oversimplified simulation environments may fail to capture real-world complexity, necessitating bridging the sim-to-real gap. It's typically addressed through domain randomization of visual and physical parameters, photo-realistic rendering with ray tracing, systematic augmentations encompassing lighting conditions, camera viewpoints, background textures, and object appearances, as well as hybrid training that combines large-scale simulation data with minimal real demonstrations.

Additional efforts have been made to minimize the sim-to-real gap: ReBot [174] proposes a real-to-sim-to-real pipeline that grounds data generation in authentic robot trajectories rather than relying solely on simulator-based policies. The method replays identical real robot trajectories across diverse simulated scenarios with varying objects, then composites the acquired simulation robot movements onto task-agnostic real-world backgrounds obtained through inpainting techniques. Similarly, R2R2R [175] synthesizes large-scale, photorealistic robot demonstrations from real-world input: a smartphone scan of objects and a single human manipulation video. The pipeline extracts 3D assets and segments them into semantic parts, tracks 6-DoF object trajectories through 4D Differentiable Part Modeling. Differently, RealMirror [176] introduces a teleoperation-simulation combined collection framework. The system uses a motion-control pipeline in simulation to control robots in the real world, with multi-level filtering mechanisms to ensure physically plausible trajectories, and a lightweight WebXR-based communication protocol that significantly reduces end-to-end latency compared to conventional frameworks. This design enables efficient real-world data acquisition through simulated interfaces. Either way, they collected large-scale, high-fidelity simulation datasets that exhibit minimal sim-to-real gap.

5.3 Internet-Scale and Cross-Domain Data Utilization

Simulation data collection effectively addresses the scalability limitations in human teleoperation demonstrations. Despite the advancement, the approach needs to build datasets from scratch and is constrained by the sim-to-real gap. This has catalyzed a shift in focus toward exploiting internet-scale and other existing data sources. This emerging approach seeks to harness the vast, diverse, yet unstructured, unannotated data repositories available online, including egocentric human videos and community-contributed robot datasets. The central challenge lies in reconciling the inherent heterogeneity of these data—spanning embodiment

TABLE 6: Representative Works on Efficient Data Collection.

Method/Model	Year	Key Innovation in Efficient Data Collection
(A) Human-in-the-loop Data Collection		
CLIP-RT [79]	2024	Enables non-expert data collection using language teleoperation and stochastic augmentation.
GCENT [173]	2025	Integrates human rewind-and-refine guidance for efficient, failure-centric data collection.
(B) Simulation Data Collection		
GeRM [85]	2024	Leverages pre-trained policies to autonomously collect mixed-quality demonstration data.
GraspVLA [50]	2025	Develops an efficient, parallelized pipeline for massive-scale synthetic grasping data generation.
cVLA [147]	2025	Leverages simulation with domain randomization to enable scalable VLA training without costly real-world data.
RoboTwin 2.0 [16]	2025	Combines automated code synthesis with multimodal feedback loops for scalable expert data generation.
ReBot [174]	2025	Employs real-to-sim-to-real pipeline combining trajectory replay with background inpainting.
R2R2R [175]	2025	Synthesizes large-scale data from smartphone scans and single demonstrations without physics simulation.
RealMirror [176]	2025	Introduces simulated teleoperation system to acquire high-quality training data without physical robots.
(C) Internet-Scale and Cross-Domain Data Utilization		
SmolVLA [20]	2025	Standardizes heterogeneous community datasets for scalable VLA pretraining.
EgoVLA [140]	2025	Leverages large-scale egocentric human videos for pretraining VLA models to overcome robot data scarcity.
RynnVLA [142]	2025	Introduces automated ego-centric video curation pipeline transferring human manipulation skills to robot learning.
Egoscaler [177]	2025	Extracts trajectories from egocentric videos without auxiliary annotations or specialized hardware.
Being-H0 [141]	2025	Introduces physical instruction tuning paradigm that pretrains on human hand videos.
MimicDreamer [178]	2025	Aligns human videos to robot domain via joint vision-viewpoint-action transformation.
EMMA [179]	2025	Enables text-controlled visual transfer of robot demonstrations.
Humanoid-VLA [145]	2025	Language-motion alignment via compositional quantization and pseudo-annotation.
(D) Self-Exploration Data Collection		
Anypos [146]	2025	Automates task-agnostic action collection via RL-based workspace coverage.
SimpleVLA-RL [13]	2025	Employs online RL to self-generate diverse trajectories from minimal human demonstrations.
DiffusionRLVLA [180]	2025	Generates synthetic demonstrations via diffusion RL replacing costly human data.
World-Env [167]	2025	Combines small, pre-trained models in a modular framework for consumer-grade GPU deployment.
VLA-RFT [181]	2025	Employs world model simulator for efficient reinforcement fine-tuning with trajectory-level rewards.
(E) Data Augmentation		
LLaRA [17]	2025	Reformats datasets into conversational instruction-response pairs with templates.
InstructVLA [157]	2025	Uses GPT-4o annotations to mitigate catastrophic forgetting.
RoboChemist [9]	2025	Fusing multi-modal prompts and failures enhances self-correction.
ReconVLA [182]	2025	Segments gaze regions for visual reconstruction pre-training.
CLIP-RT [79]	2025	Stochastically drives robot to novel states beyond demonstrations.
ERMV [183]	2025	Propagates frame edits across multi-view timesteps consistently.

disparities, viewpoint discrepancies, and action space misalignments—and transforming them into formats amenable to VLA training. Towards this, researchers have developed several distinct yet complementary strategies.

One strategy focuses on curating and standardizing the noisy, heterogeneous data that already exists within the robotics community. The SmolVLA [20] paper exemplifies this community-driven approach by aggregating numerous small-scale datasets from platforms like Hugging Face. To tackle the inherent inconsistencies, it employs a VLM to automatically generate clean, consistent task descriptions from noisy original labels and manually maps diverse camera viewpoints into a standardized format. This curation-heavy strategy demonstrates that even with a dataset an order of magnitude smaller than state-of-the-art, high performance can be achieved by maximizing the quality and diversity of real-world, albeit messy, data.

While curating community-contributed robot datasets partially enhances efficiency, the most abundant data source, however, remains human videos, more commonly, egocentric videos. In this pathway, the core obstacle lies in bridging the human-robot embodiment gap.

EgoVLA [140] pioneers this egocentric paradigm: it introduces the foundational concept of treating humans as

a form of robot, establishing egocentric human videos as a viable training modality for VLAs. The work constructs a large-scale Ego-Centric Human Manipulation Dataset aggregating skill-rich sequences from multiple sources. Identically, EgoVLA [140] is pretrained on this heterogeneous human dataset. It reveals the potential of leveraging abundant, unstructured human egocentric videos to achieve superior generalization. Building upon EgoVLA’s conceptual foundation, RynnVLA-001 [142] addresses the practical challenge of massive egocentric manipulation video acquisition by establishing an automated, multi-stage data curation pipeline. The pipeline leverages pose estimation to identify egocentric perspectives through the absence of facial landmarks and the presence of hand keypoints. Following the acquisition of web-scale egocentric videos, EgoScaler [177] narrows the human-robot embodiment gap by transforming raw, unstructured visual data into structured robotic action representations. The work introduces an automated pipeline that extracts 6-DoF object trajectories directly from egocentric videos without manual annotation, thereby converting passive visual observations into actionable manipulation sequences suitable for robot policy learning. Progressively, Being-H0 [141] elevates action representation from coarse object trajectories to fine-grained hand poses, addressing the

precision and standardization requirements for dexterous manipulation tasks.

In a more advanced manner, MimicDreamer [178] confronts the human-robot embodiment gap by translating human demonstration videos into synthetic sequences that conform to robotic appearance and dynamics through video diffusion models, directly resolving visual and kinematic discrepancies. The framework canonicalizes egocentric videos through stabilization and inpainting, then employs constrained inverse kinematics to map human wrist trajectories into robot joint configurations, and critically, deploys a video diffusion model to synthesize photorealistic robot demonstration videos that conform to robotic embodiment constraints. Consequently, the training data observed by VLA models becomes visually identical to what they encounter during real-world task execution, thereby resolving the embodiment gap between human demonstrations and robotic operation. DreamTransfer [179] also introduces a diffusion Transformer framework for generating photorealistic, multi-view consistent robot manipulation videos. The method leverages pretrained diffusion models to jointly encode synchronized multi-view depths and text prompts, enabling text-controlled visual transfer of real or simulated demonstrations.

However, Internet-scale human videos often lack language annotations and egocentric perspectives. To address this, HumanoidVLA [145] uses a generalizable methodology that leverages third-person human-motion videos. By decomposing body poses into part-specific tokens and applying temporal and spatial perturbations with corresponding instructional prompts, the approach transforms raw third-person videos into structured training signals without manual annotation. This framework expands the scope of usable human data beyond first-person videos and further alleviates data scarcity through scalable self-supervision.

5.4 Self-Exploration Data Collection

While the aforementioned paradigms enhance data collection efficiency through diverse mechanisms (*e.g.*, active collection and cross-domain transformation), they fundamentally remain constrained by passive dependence on generated trajectory data. To transcend this limitation, an emerging research paradigm shifts from passive data reception toward autonomous exploration, wherein agents actively generate valuable training data through environment interaction. This paradigm fundamentally reframes data collection from a human-intensive bottleneck into an agent-driven, self-improving process, positioning self-exploration as a pivotal enabler for scalable and efficient data collection.

Before addressing task-specific learning, a fundamental question emerges: Does the robot possess comprehensive awareness of its physical capability boundaries? Without a systematic exploration of its kinematic reachable space, downstream task learning risks operating on incomplete behavioral priors. AnyPos [146] tackles this foundational challenge through ATARA, a self-supervised framework that decouples data generation from task semantics by leveraging RL-driven policies to orchestrate efficient and uniform coverage of the robot’s end-effector workspace, autonomously synthesizing large-scale task-agnostic (image, action) datasets. This approach mitigates limitations

in naive random exploration—sparse coverage, motion redundancy, and frequent self-collisions—thereby establishing a reusable, unbiased, and physically grounded kinematic foundation for all subsequent task learning, answering “what the robot can do” before prescribing “what it should do”.

Task-agnostic exploration [146] equips VLAs with a “kinematic prior” that characterizes their physical capabilities. However, executing specific user instructions demands task-relevant behavioral sequences with purposeful intent. Upon this, online reinforcement learning (RL) emerging as the predominant pathway that enables agents to “learn while collecting” through direct environment interaction.

SimpleVLA-RL [13] pioneers the demonstration of self-driven data collection through online RL with only minimal human demonstrations as “seeds”. The framework transforms the VLA itself into a high-quality trajectory data generator through a “generate-evaluate-optimize” cycle: generating diverse trajectories via interactive rollouts with stochastic action sampling, filtering them with binary success rewards, and retaining successful executions as training data while simultaneously optimizing the policy. Nevertheless, this direct RL fine-tuning approach remains constrained by the policy model’s representational capacity—for complex, multimodal behaviors, standard policies may converge to local optima or produce suboptimal averaging actions, limiting exploration breadth and trajectory quality. To address this challenge, Yang [180] proposes a more sophisticated solution wherein the policy architecture is replaced with the highly expressive diffusion model. The diffusion model’s powerful generative capacity enables superior capture of multi-modal distributions inherent in human demonstrations while generating smoother, more consistent, and higher-quality near-optimal trajectories during RL exploration. Crucially, this methodology produces a synthetic dataset that surpasses the quality of original human demonstrations.

While online RL demonstrates notable efficacy, it needs extensive interaction with either physical environments or high-fidelity physics simulators. Consequently, the next evolutionary trajectory naturally migrates exploration from costly physical domains to low-cost virtual worlds. World-Env [167] provides a direct solution by leveraging existing high-fidelity simulators (*e.g.*, LIBERO): starting from minimal expert demonstrations, it deploys VLA policies to explore within the simulator through controlled stochasticity, thereby economically augmenting diverse interaction data. However, this approach remains constrained by pre-constructed simulators that may fail to generalize across novel environments. In such cases, VLA-RFT [181] advances this methodology to its extreme by learning a data-driven world model directly from offline interaction datasets, eliminating reliance on high-fidelity simulators. This learned world model serves as a controllable simulator that captures the diversity of real-world interactions, enabling the VLA policy to undergo reinforcement fine-tuning entirely within a synthetic environment through massively parallel rollouts of predicted visual trajectories. The approach fundamentally transitions data collection from passive accumulation to active generation.

5.5 Data Augmentation

Data augmentation can also be positioned as strategies of efficient data collection, which is achieved by maximizing the utility and diversity of existing data, transforming collected trajectories into richer, more varied training signals.

A prominent approach involves enriching linguistic and semantic annotations. LLaRA [17] pioneered the automated reformatting of existing behavior cloning datasets into conversational instruction-response pairs using templates, while also defining auxiliary tasks for self-supervised enhancement. Building significantly on this, InstructVLA [157] utilizes advanced LLMs (GPT-4o) to curate diverse, hierarchical annotations—including scene captions, QA pairs, and command rewriting—from large manipulation datasets, specifically to mitigate the catastrophic forgetting of pre-trained VLM capabilities during fine-tuning. This strategy of using models for annotation is also seen in RoboChemist [9], which employs LLMs to diversify language instructions and VLMs to generate automated visual prompts (*e.g.*, labeling grasp points) to ensure safety compliance. Other methods generate novel training objectives from existing data; ReconVLA [182], for instance, fine-tunes Grounding DINO on robotic datasets to automatically segment interaction-relevant “gaze regions”, thereby constructing a large-scale pre-training dataset dedicated to visual reconstruction.

Augmentation can also target the trajectory, state, and temporal dimensions. CLIP-RT [79] introduced Stochastic Trajectory Augmentation (STA), which stochastically drives the robot into novel states beyond the expert demonstrations and applies simple heuristics for automatic labeling. RoboChemist [9] enhances robustness differently, by explicitly injecting teleoperated failure scenarios and retry attempts into the training data, improving the model’s capacity for self-correction.

Finally, several methods achieve augmentation by directly manipulating the visual modality. Addressing existing 4D data, ERMV [183] proposes a framework that applies targeted edits (*e.g.*, inpainting) to an initial frame and propagates these changes consistently across all views and timesteps using Epipolar Motion-Aware Attention, generating new visual sequences paired with the original, unmodified actions.

5.6 Discussion

5.6.1 Innovations

The trajectory of VLA data acquisition has pivoted from labor-intensive teleoperation toward computation-driven regimes that effectively decouple dataset expansion from human temporal constraints. Simulation frameworks and autonomous exploration now facilitate parallelized generation, substituting manual effort with computational throughput to achieve unprecedented scale. This shift is further augmented by cross-domain manifold alignment, which leverages internet-scale egocentric videos as kinematic proxies through advanced retargeting and generative synthesis. Consequently, the integration of generative augmentation transforms the data pipeline from passive accumulation into an active process that maximizes the information density of sparse expert signals.

5.6.2 Limitations

Notwithstanding these advances, substantial bottlenecks persist in reconciling high-throughput data with real-world physical utility. The reliance on synthetic and cross-modal sources introduces inherent sim-to-real and embodiment discrepancies, where misalignments in contact dynamics and kinematic topology degrade policy transferability. While autonomous exploration alleviates data scarcity, it necessitates a delicate balance between sample efficiency and safety, often constrained by world models prone to physics hallucinations. Ultimately, the lack of precise action labels in internet-scale datasets imposes a fundamental ceiling on control fidelity, which necessitates continued reliance on in-domain expert supervision for high-precision tasks.

6 APPLICATIONS

The pursuit of efficiency in Vision-Language-Action (VLA) models is not an end in itself, but a critical enabler for their deployment in the physical world. The techniques outlined in preceding chapters—from streamlined architectures to efficient training paradigms and data curation strategies—culminate in their practical utility across a spectrum of real-world robotic applications. This chapter delineates some representative applications, demonstrating how efficient VLAs are transforming domains that demand real-time response, operational robustness, and computational frugality.

6.1 Intelligent Vehicles and Autonomous Driving

Intelligent vehicles operate under one of the most stringent sets of constraints for any embodied AI system. They must process high-dimensional sensor data in real-time, understand complex traffic scenarios, and execute safe control commands—all within the severe computational and power budgets of a mobile platform. Bulky models are fundamentally unsuitable for this domain. Efficient VLAs address this by enabling direct, end-to-end mapping from sensory inputs to driving actions with minimal latency. Compressed and optimized models can be deployed on automotive-grade hardware, facilitating nuanced driving behavior based on multimodal inputs, such as interpreting a traffic officer’s gesture or responding to a complex verbal navigational command. Recent research efforts [184], [185], [186], [187] have echoed this technical imperative. The computational leanness of these models is not merely a convenience; it is a prerequisite for safety and reliability, allowing for rapid inference that keeps pace with the dynamic driving environment.

6.2 Family Robotics and Smart Home

The home environment presents a unique set of challenges centered on privacy, persistent operation, and natural human-robot interaction. Cloud-dependent architectures raise significant privacy concerns and introduce latency that breaks the flow of interaction. Efficient VLAs are pivotal in shifting the intelligence to the edge, directly onto the robot. Lightweight models empower service robots to comprehend and execute open-ended commands like “tidy up the living room” entirely offline, ensuring user data never leaves

the device. This on-board processing capability, coupled with low power consumption, enables robots to offer assistance over extended periods without frequent recharging. Furthermore, the reduced inference latency is crucial for sustaining engaging and safe conversational interactions, making the robot a responsive and seamless part of the domestic fabric.

6.3 Industrial Manufacturing and Logistics

Industrial settings demand precision, high throughput, and scalability. The vision of large-scale, collaborative robot fleets hinges on the cost-effectiveness and computational efficiency of the underlying AI models. Efficient VLAs are the cornerstone of this vision. By dramatically reducing the parameter counts and computational overhead, these models make it economically viable to deploy advanced intelligence across hundreds of manipulators and autonomous guided vehicles (AGVs). They enable real-time visual recognition for precise part selection and assembly, while their low latency ensures operational cycles meet demanding production line tempos. Beyond single-task execution, efficient VLAs facilitate quick redeployment through natural language instructions, allowing a single robot to perform multiple functions—from “pick and place component A” to “inspect the final product for defects”—thereby enhancing overall manufacturing flexibility and agility. For example, CIPHER [7] efficiently switches between tasks in the 3D printing industry—from tuning extrusion parameters for layer precision to conducting visual defect scans—enabling seamless, multi-role adaptation through natural language without hardware changes.

6.4 Medical Assistive Robotics

In medical and assistive contexts, the imperatives of precision, data security, and personalized adaptation converge. Surgical and rehabilitation robots require exquisitely fine-grained control, which in turn demands real-time sensorimotor processing. The sensitive nature of health information mandates that patient data be processed locally, precluding the use of cloud-based models. Efficient VLAs are uniquely suited to meet these dual challenges. Their optimized architecture allows for the low-latency, high-precision control loops essential for assisting in delicate procedures. By operating entirely on-premise, they guarantee the confidentiality and privacy of patient data. Moreover, the data-efficient nature of these models, often leveraging powerful pre-trained backbones, permits effective fine-tuning with limited patient-specific datasets. This enables a new level of personalization, where assistive devices can quickly adapt to the unique physiology and needs of an individual user, paving the way for more accessible and effective personalized care.

7 CHALLENGES AND FUTURE WORKS

In this chapter, we delineate the principal challenges impeding the maturation of efficient VLAs and propose forward-looking research trajectories. Anchored in the taxonomy elucidated earlier, encompassing efficient model design, efficient training and efficient data collection, we dissect

these facets across Model, Training, and Data dimensions. This structured exposition aims to catalyze advancements toward scalable embodied AI, mitigating computational exigencies while amplifying performance in robotic manipulation and beyond.

7.1 Challenges

Notwithstanding strides in efficient VLAs, enduring impediments curtail their scalability and robustness. These arise from intrinsic trade-offs in multimodal integration, where vision, language, and action modalities demand harmonious yet parsimonious orchestration, often yielding sub-optimal generalization in dynamic settings.

7.1.1 Model: Balancing Compactness and Expressivity

Efficient VLA design confronts a fundamental tension between architectural parsimony and representational richness, where aggressive compression sacrifices fidelity in capturing fine-grained spatiotemporal dynamics. Inference acceleration in efficient VLA design inevitably trades model capability for speed, undermining cross-embodiment robustness. Hierarchical or modular paradigms, while promising scalability, introduce routing overheads that undermine real-time deployability, particularly on resource-constrained hardware. Ultimately, the pursuit of sub-billion-parameter efficiency must contend with emergent degradation in long-horizon reasoning and adaptation to unseen tasks, which may hinder seamless transition from simulation to physical worlds.

7.1.2 Training: Scalability versus Stability Trade-offs

Training efficient VLAs is beset by the dual imperatives of computational frugality and convergence reliability. Pre-training efficient VLAs, despite leveraging frozen vision-language backbones, still incurs substantial compute for action head alignment and risks brittle generalization under embodiment shifts in downstream tasks. Post-training adaptation, especially via reinforcement learning, grapples with high-variance gradients and reward sparsity, exacerbating instability in policy refinement. Action representation compression, though expedient, distorts continuous kinematics, impeding transfer across embodiments. These trade-offs fundamentally hinder scalable, reproducible training pipelines, curtailing widespread adoption across heterogeneous robotic platforms.

7.1.3 Data: Quality, Diversity, and Accessibility Barriers

Data remains the bedrock and bottleneck of VLA efficacy, plagued by scarcity of high-fidelity, task-diverse trajectories. Human-sourced collections suffer from prohibitive costs, while synthetic alternatives falter in physical realism, yielding persistent sim-to-real gaps. Augmentation strategies, though volume-enhancing, risk injecting distributional biases that erode generalization. Self-supervised or exploration-driven paradigms generate voluminous but noisy signals, necessitating costly curation. The absence of standardized, ethically sourced, cross-domain repositories stifles reproducible progress and equitable access, intensifying disparities in embodied AI development.

7.2 Future Works

To surmount these obstacles, emergent directions should transcend incremental gains, embracing paradigm shifts that redefine efficiency as holistic system-level optimization. We advocate integrative, interdisciplinary approaches—spanning architecture, training theory, and data ecosystems—to forge VLAs that are not merely lightweight but fundamentally scalable, adaptive, and deployable across the spectrum of embodied intelligence.

7.2.1 Model: Toward Adaptive, Embodiment-Agnostic Architectures

Future VLA designs must evolve toward intrinsic adaptability, dynamically modulating complexity in response to task and hardware contexts. Dynamic token pruning with context-aware routing could modulate computational paths on-the-fly, preserving critical spatiotemporal details while achieving sub-linear scaling across embodiments. Modality-agnostic backbones, coupled with token orchestration, promise unified efficiency across vision, language, and action streams. Hardware-software co-design, leveraging architectural optimization of computing platforms, may shatter current latency barriers, enabling edge-native VLAs that seamlessly span consumer devices to industrial manipulators.

7.2.2 Training: Scalable, Resilient Learning Paradigms

Training regimes should pivot toward decentralized, continual, and theoretically grounded protocols. Federated paradigms, augmented with differential privacy, could harness distributed robotic agents for lifelong learning, amortizing costs while enriching data diversity. Physics-informed objectives, integrated into pre-training, may enforce kinematic consistency, bridging simulation and reality at the optimization level. Meta-learning and curriculum strategies could instill rapid adaptation, minimizing fine-tuning overhead. Ultimately, training must be reimaged as a closed-loop, self-improving process—leveraging online interaction to refine models, rendering efficiency an emergent property of deployment.

7.2.3 Data: Self-sustaining Generative Ecosystems

Data infrastructures demand transformation into generative, self-sustaining ecosystems. Diffusion-guided synthesis, conditioned on physical priors and linguistic intent, could produce infinite, verifiable trajectories from minimal seeds. Further reduction of the sim-to-real gap, *e.g.*, through the embedding of the laws of physics, can improve the reliability of simulation data and reduce the dependence on high-cost real data. Multi-agent, curiosity-driven exploration in shared virtual worlds may yield emergent task diversity, supplanting human teleoperation. Such ecosystems would not merely feed models but co-evolve with them, establishing a virtuous cycle where data quality, model capability, and real-world impact recursively amplify.

8 CONCLUSION

This survey systematically charts the emergent field of **Efficient VLAs**, providing the first unified taxonomy focusing

on the model-training-data loop to consolidate the fragmented efforts addressing their prohibitive computational and data bottlenecks. We demonstrate that current solutions converge upon three interdependent pillars, including **Efficient Model Design**, **Efficient Training**, and **Efficient Data Collection**. Our synthesis uncovers the intrinsic trade-offs between model compactness and multimodal expressivity, providing a rigorous framework to navigate the systemic tensions that characterize this design space. The challenges and future directions delineated are a direct consequence of this synthesis, charting a necessary roadmap that moves beyond isolated optimizations toward adaptive, co-designed systems. This work thus serves to catalyze this transition, accelerating the vital shift from resource-bound prototypes to truly ubiquitous physical-world intelligence.

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9 BIOGRAPHY SECTION



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